

BFS Traversal

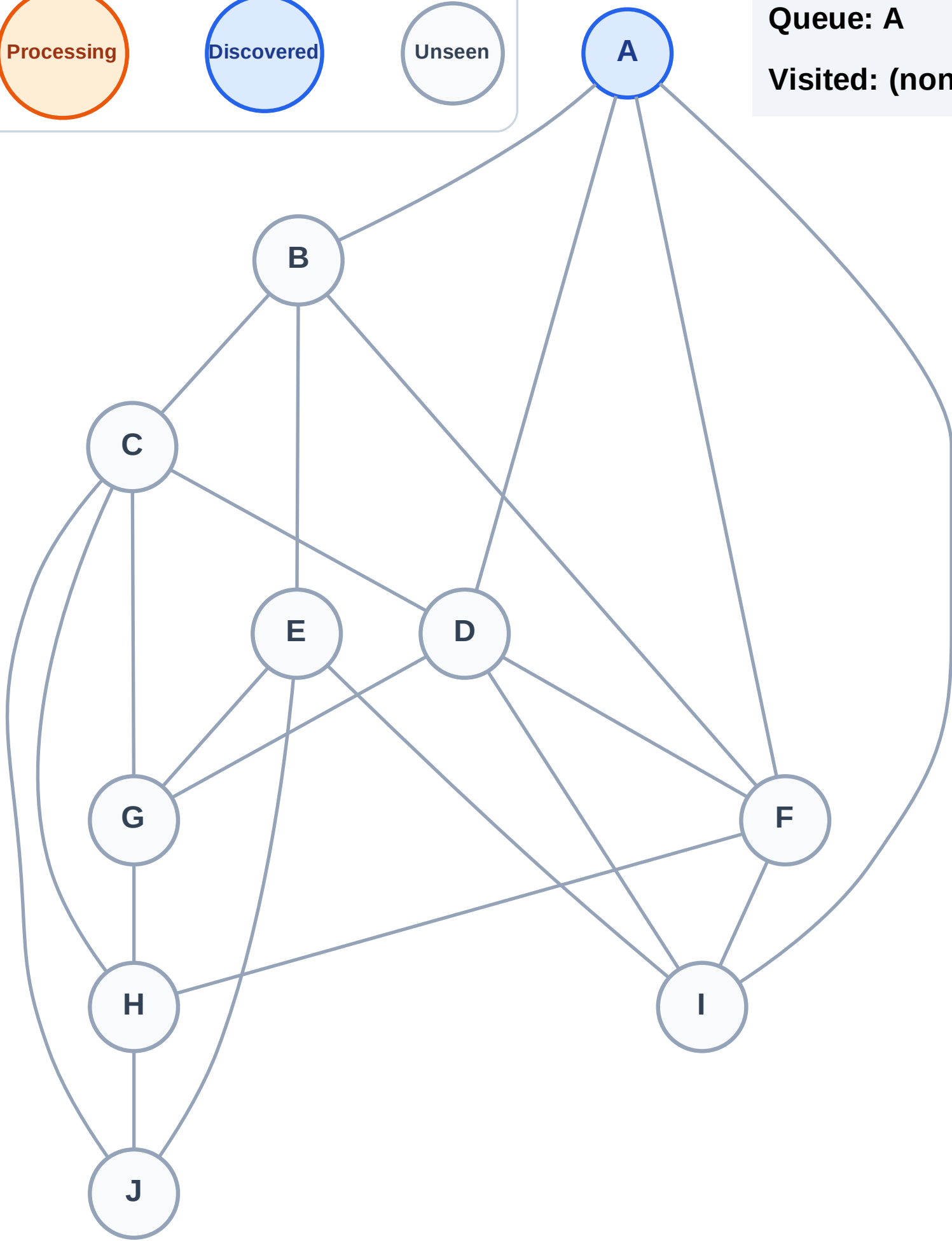
Step 1: Start at A

Legend



Queue: A

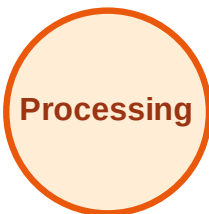
Visited: (none)



BFS Traversal

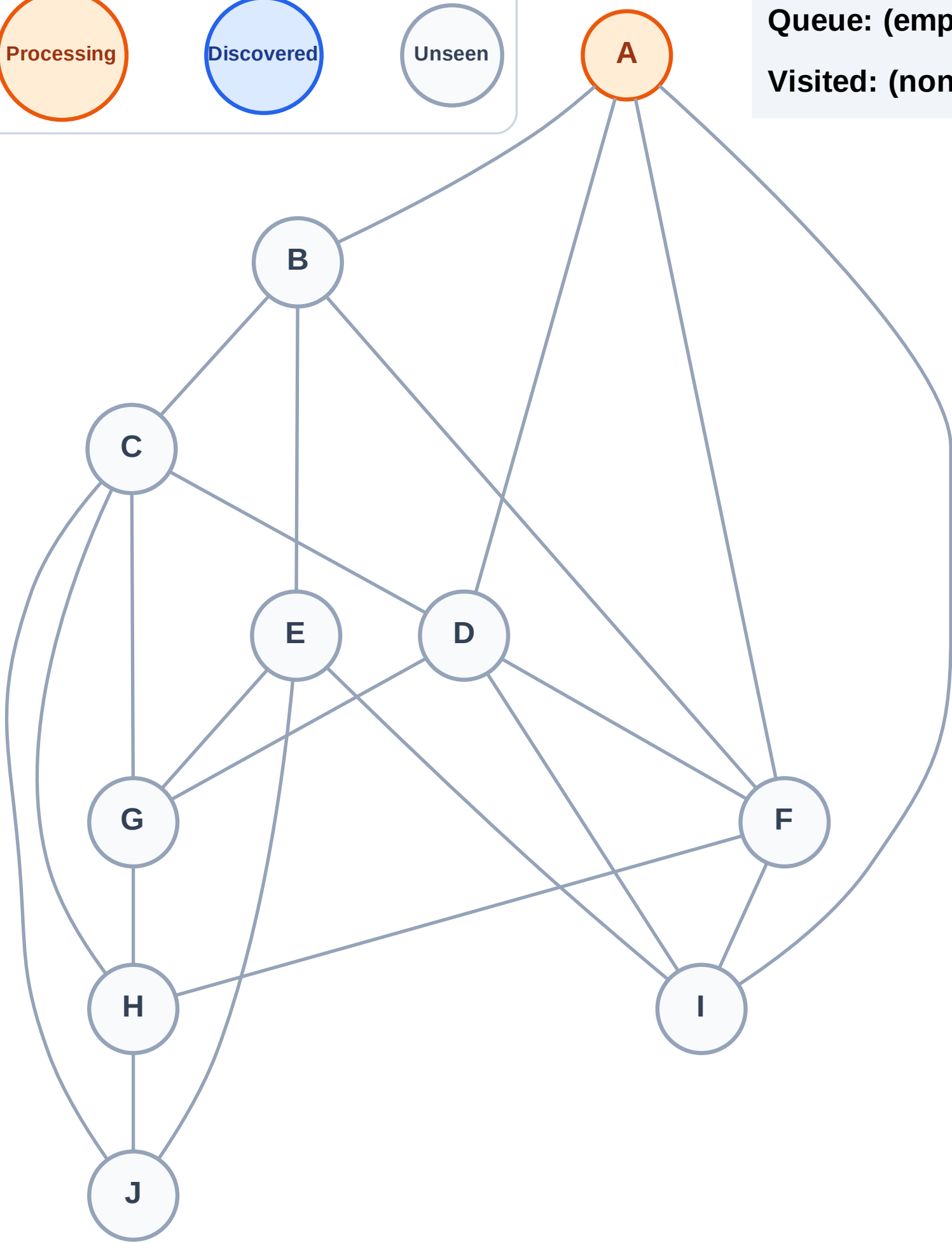
Step 2: Process node A

Legend



Queue: (empty)

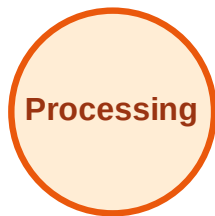
Visited: (none)



BFS Traversal

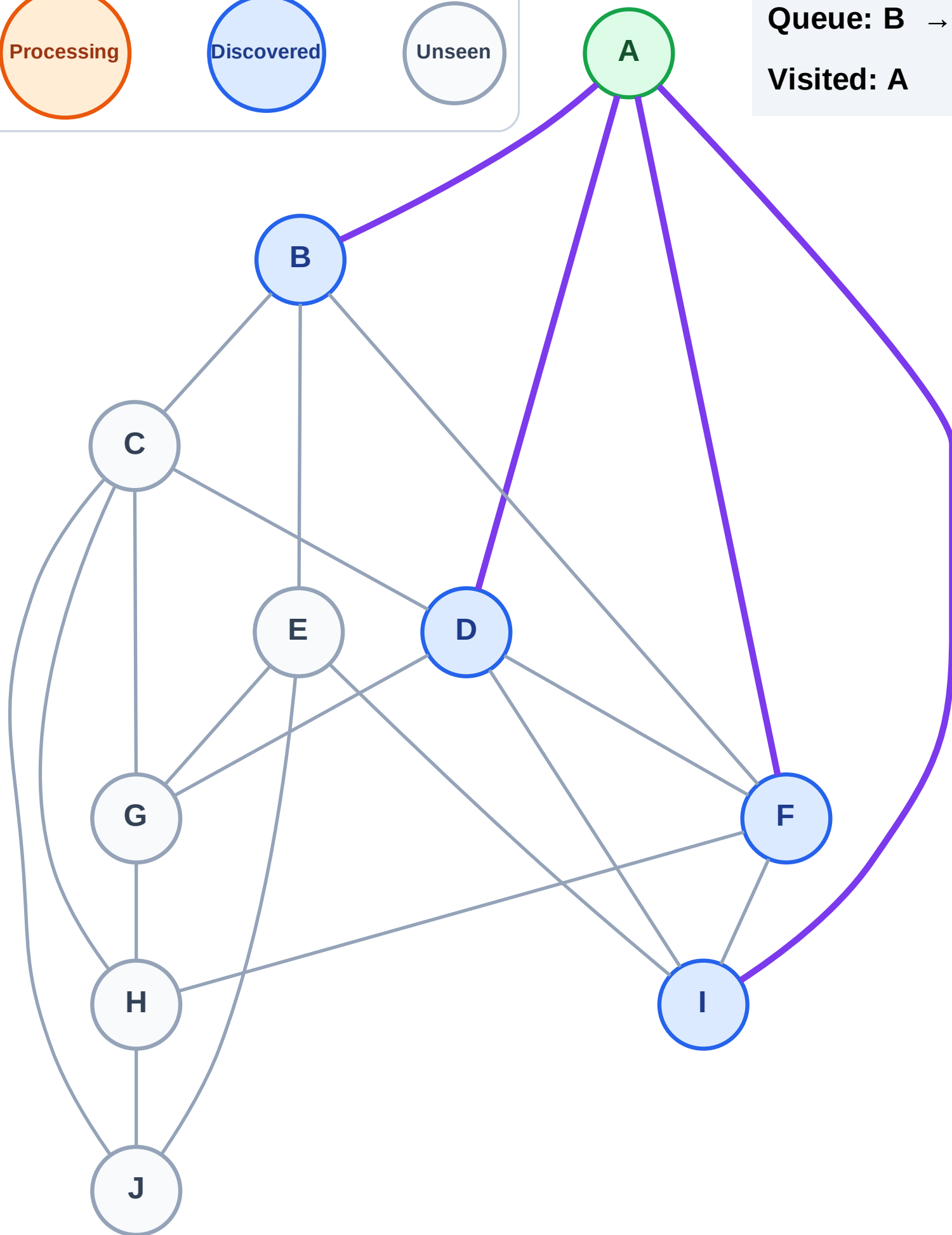
Step 3: Discover B, D, F, I from A

Legend



Queue: B → D → F → I

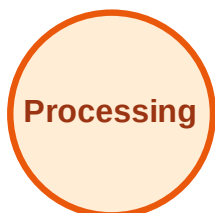
Visited: A



BFS Traversal

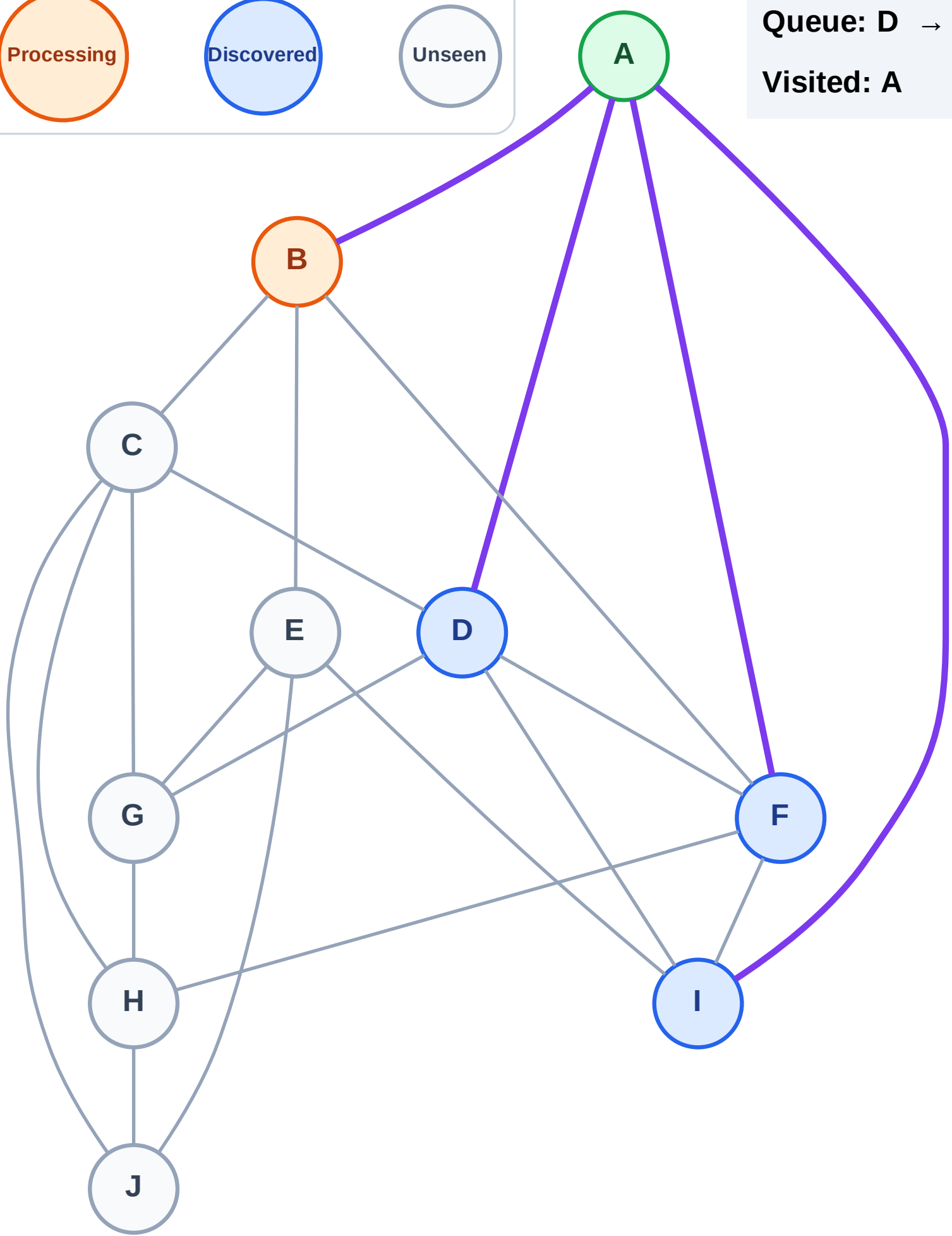
Step 4: Process node B

Legend



Queue: D → F → I

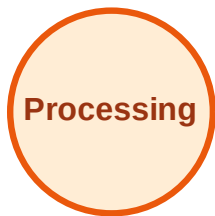
Visited: A



BFS Traversal

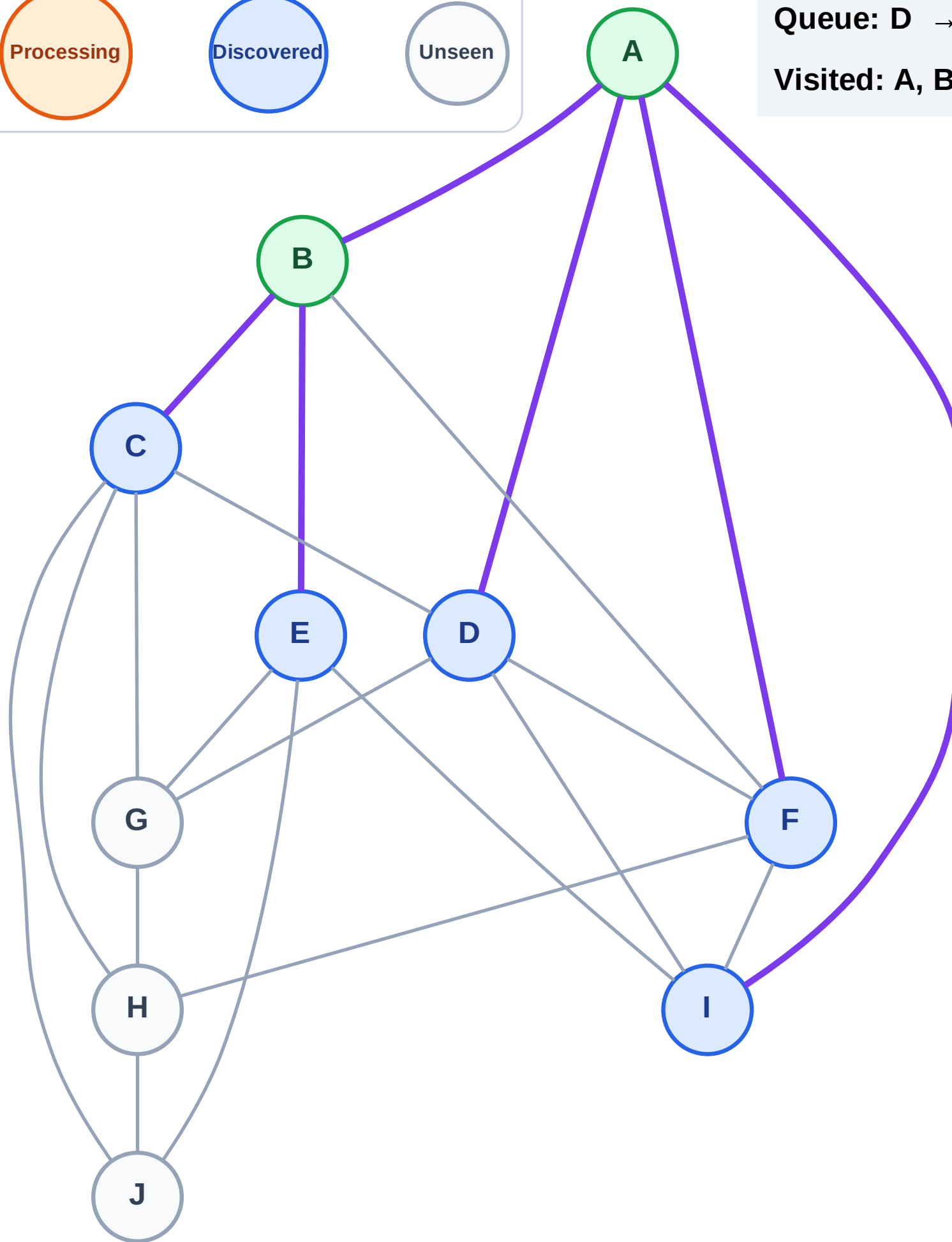
Step 5: Discover C, E from B

Legend



Queue: D → F → I → C → E

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

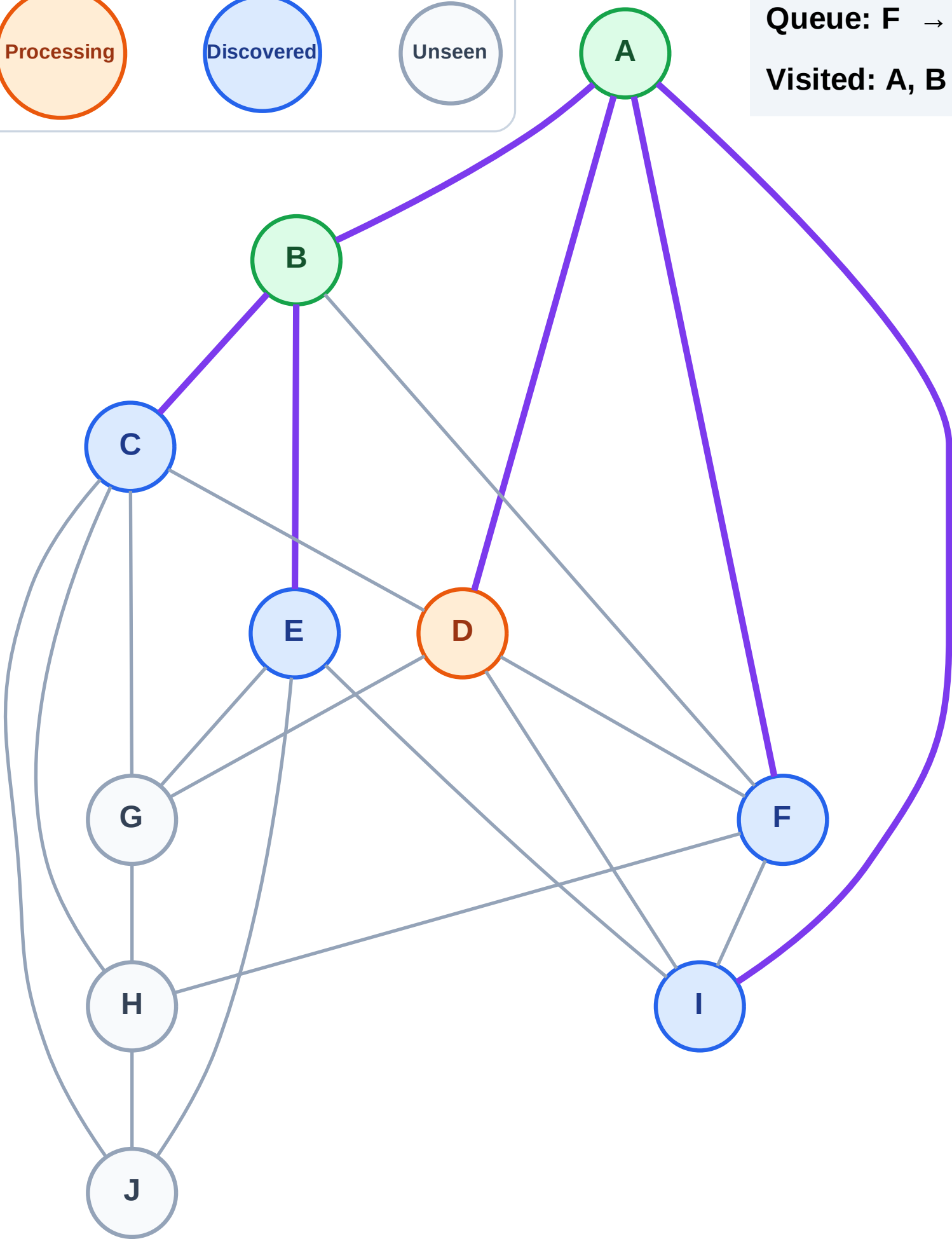
Processing

Discovered

Unseen

Queue: F → I → C → E

Visited: A, B



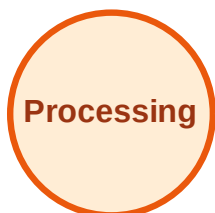
BFS Traversal

Step 7: Discover G from D

Legend



Complete



Processing



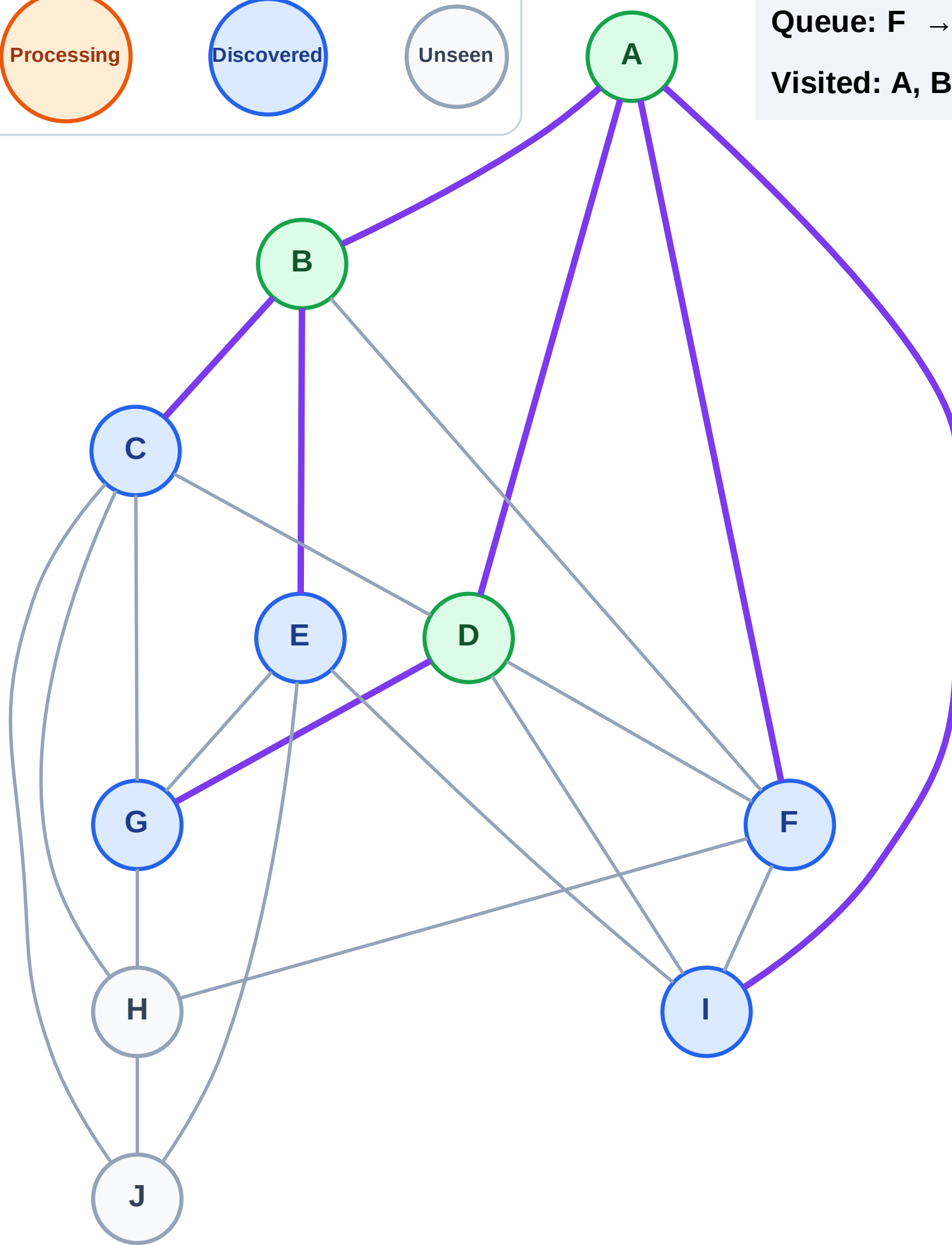
Discovered



Unseen

Queue: F → I → C → E → G

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

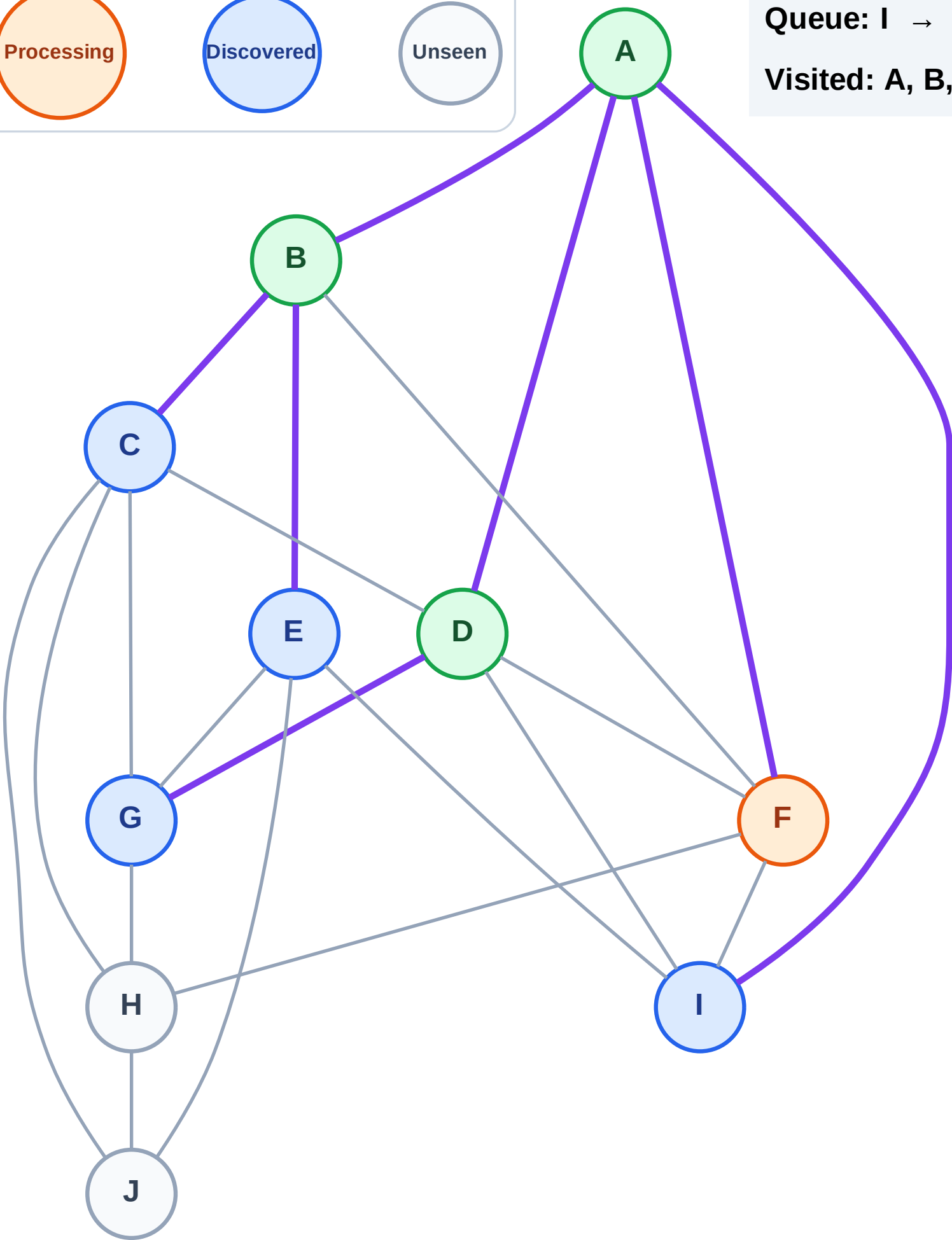
Processing

Discovered

Unseen

Queue: I → C → E → G

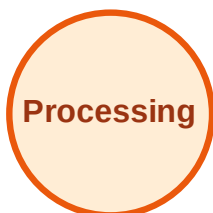
Visited: A, B, D



BFS Traversal

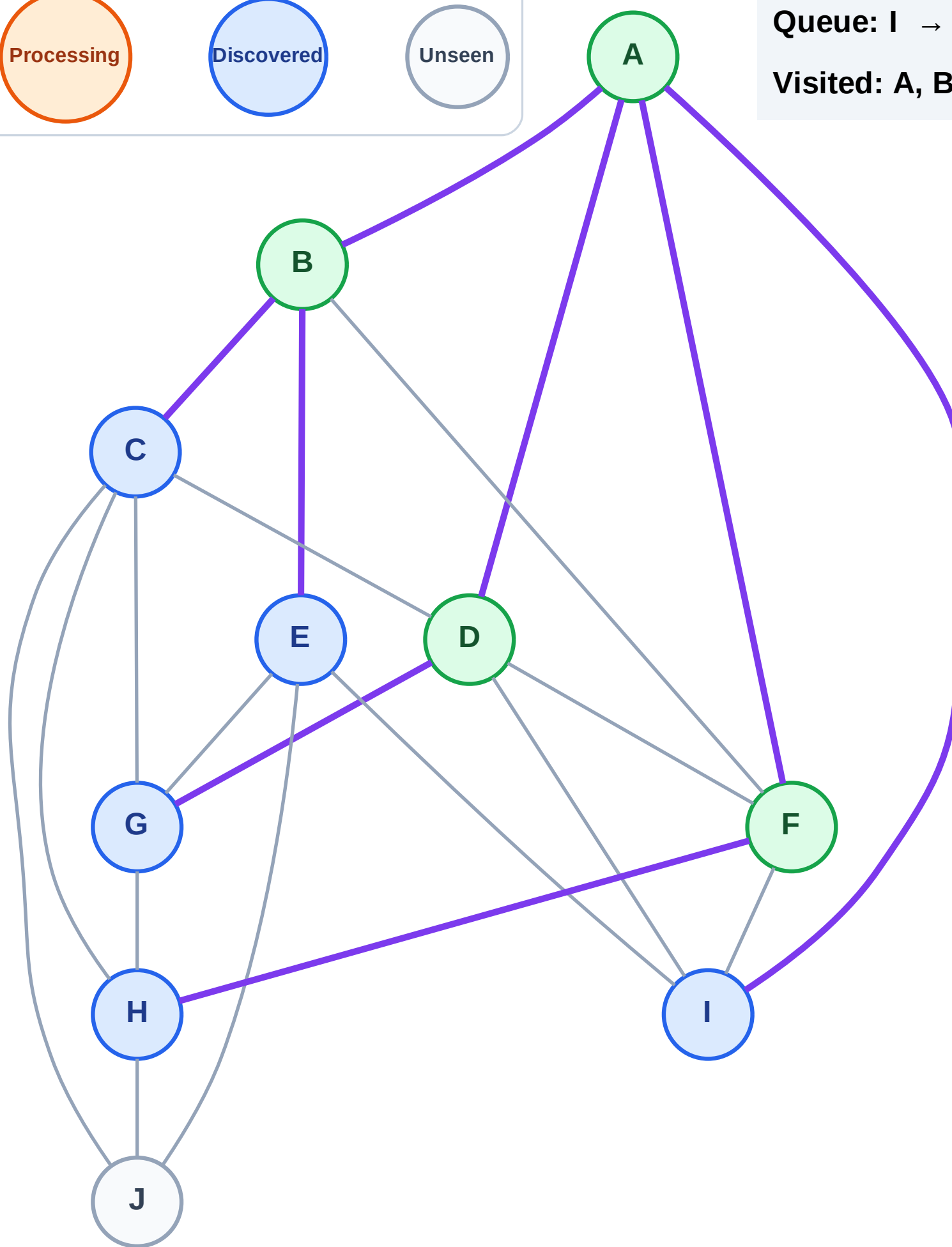
Step 9: Discover H from F

Legend



Queue: I → C → E → G → H

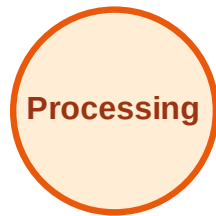
Visited: A, B, D, F



BFS Traversal

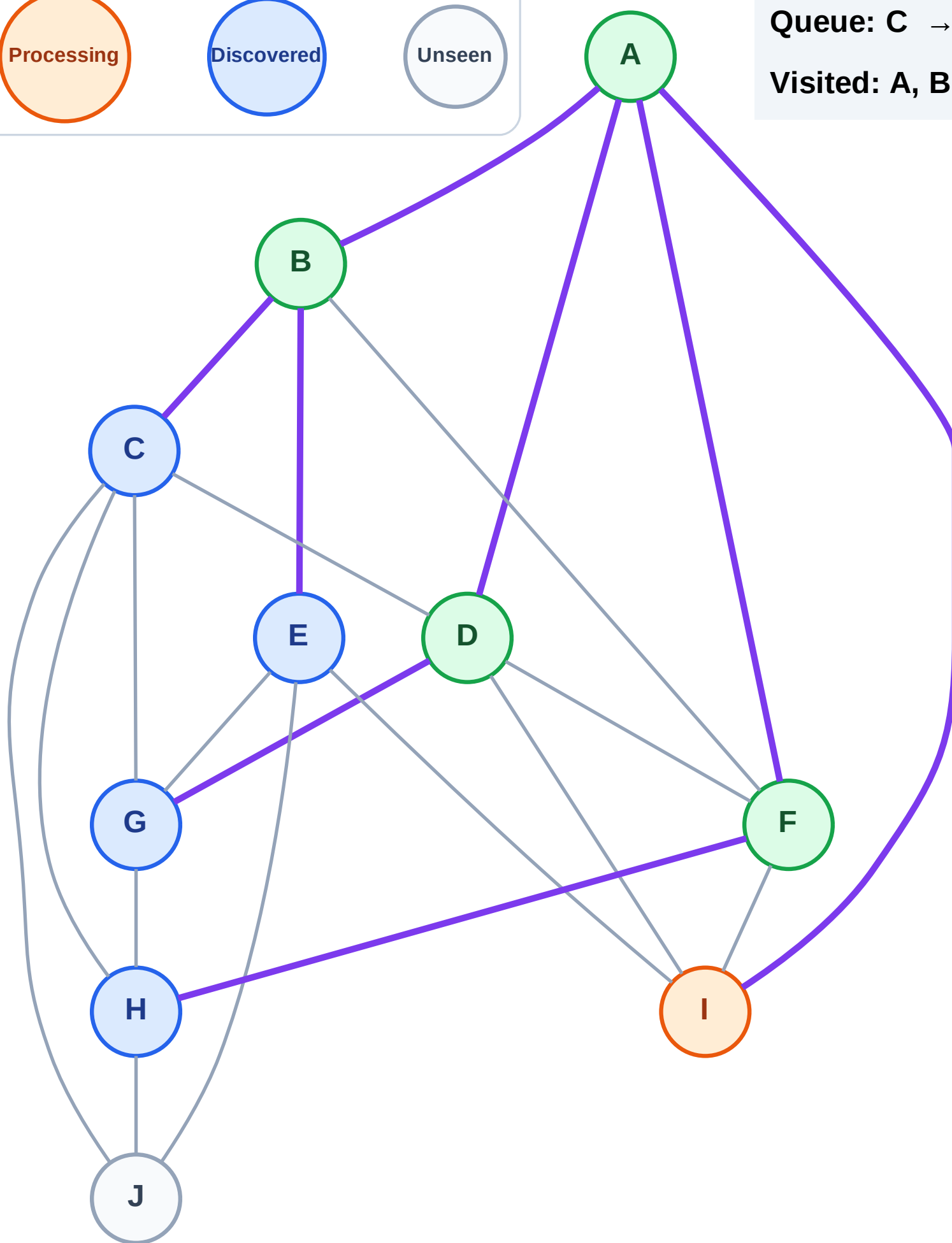
Step 10: Process node I

Legend



Queue: C → E → G → H

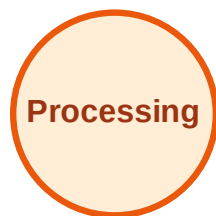
Visited: A, B, D, F



BFS Traversal

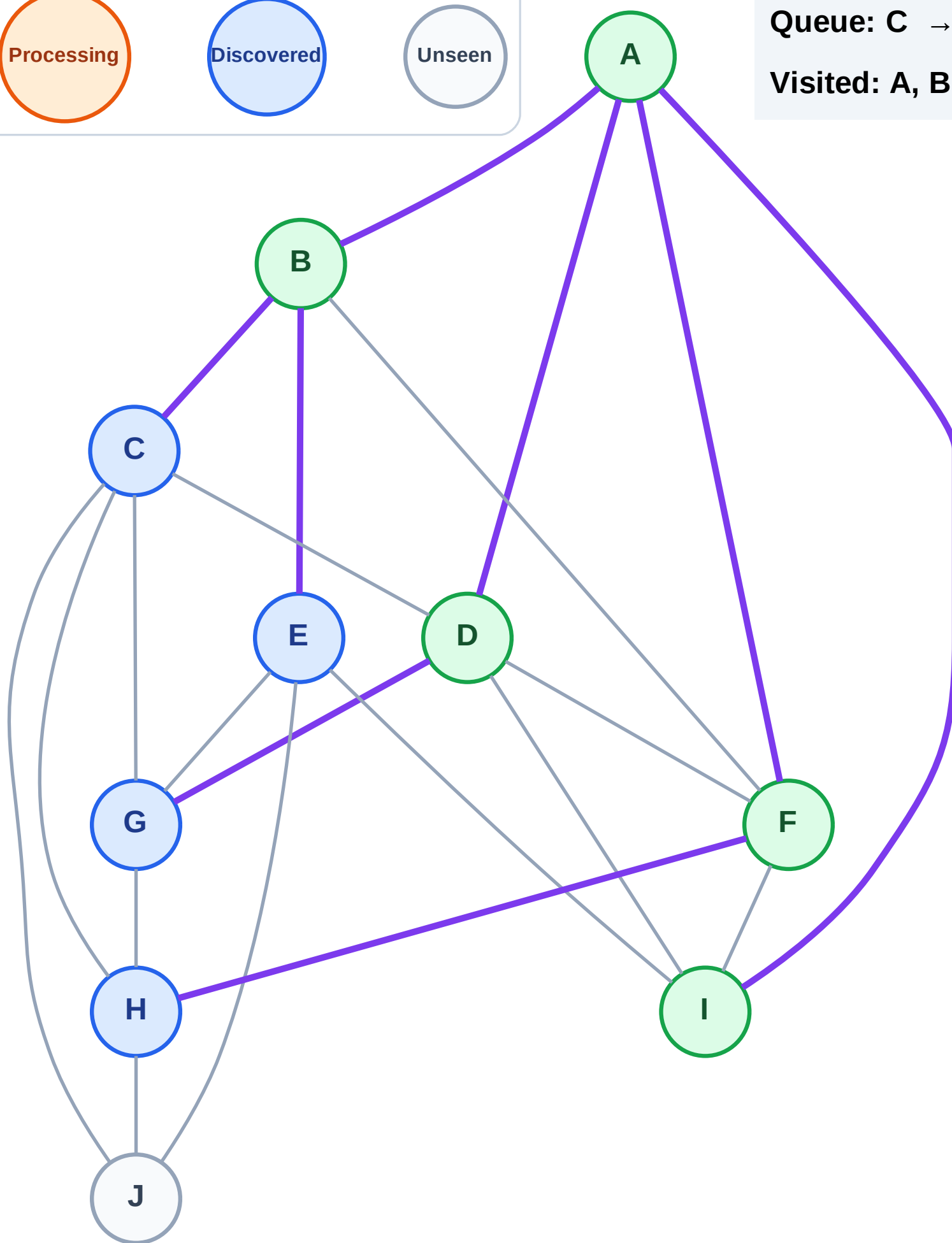
Step 11: No new neighbors from I

Legend



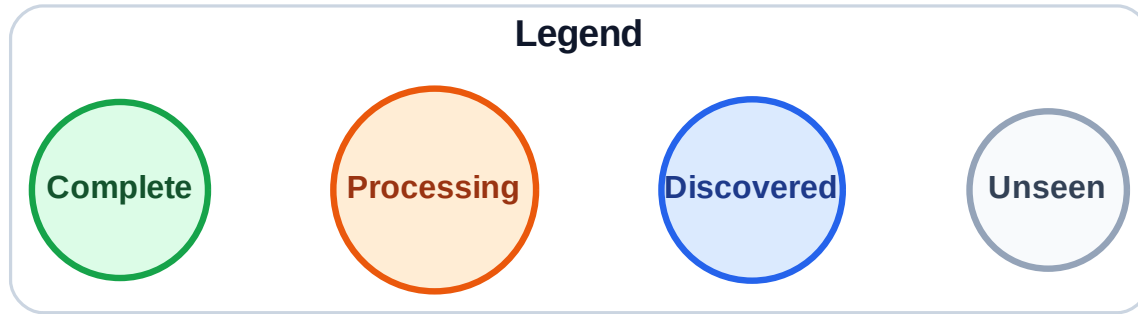
Queue: C → E → G → H

Visited: A, B, D, F, I

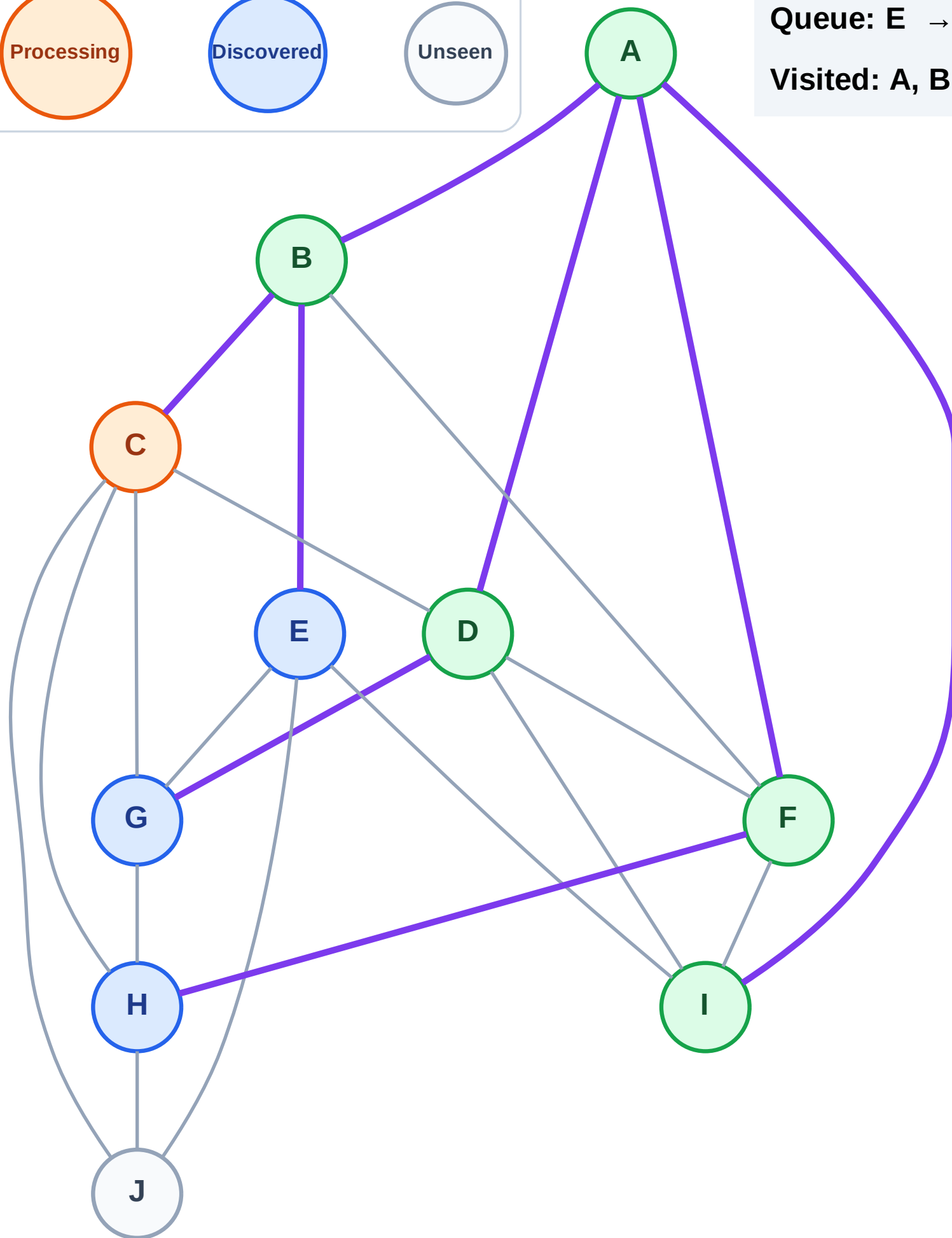


BFS Traversal

Step 12: Process node C



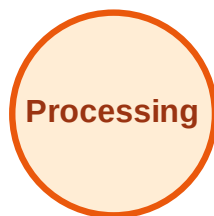
Queue: E → G → H
Visited: A, B, D, F, I



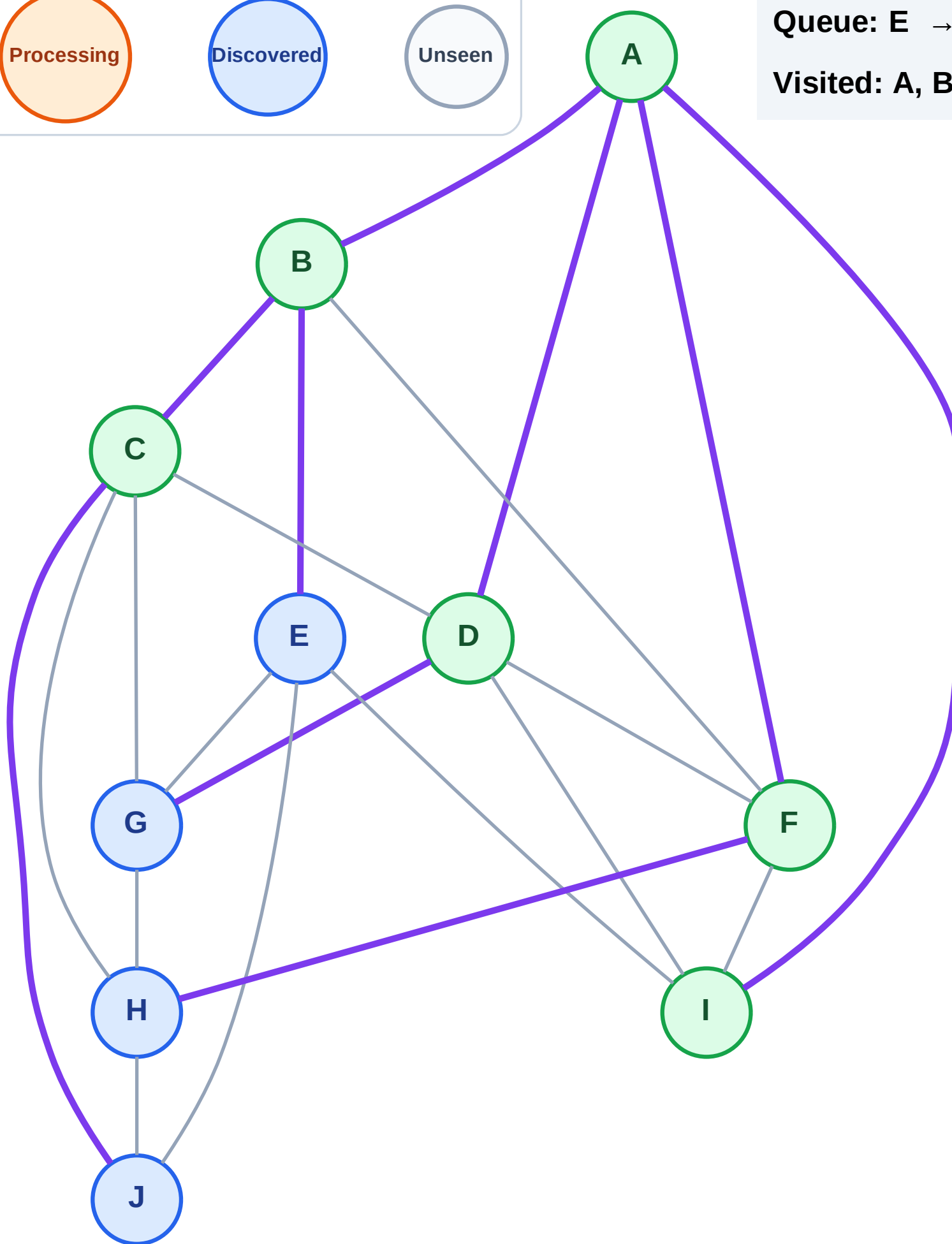
BFS Traversal

Step 13: Discover J from C

Legend



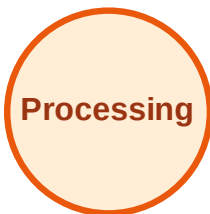
Queue: E → G → H → J
Visited: A, B, C, D, F, I



BFS Traversal

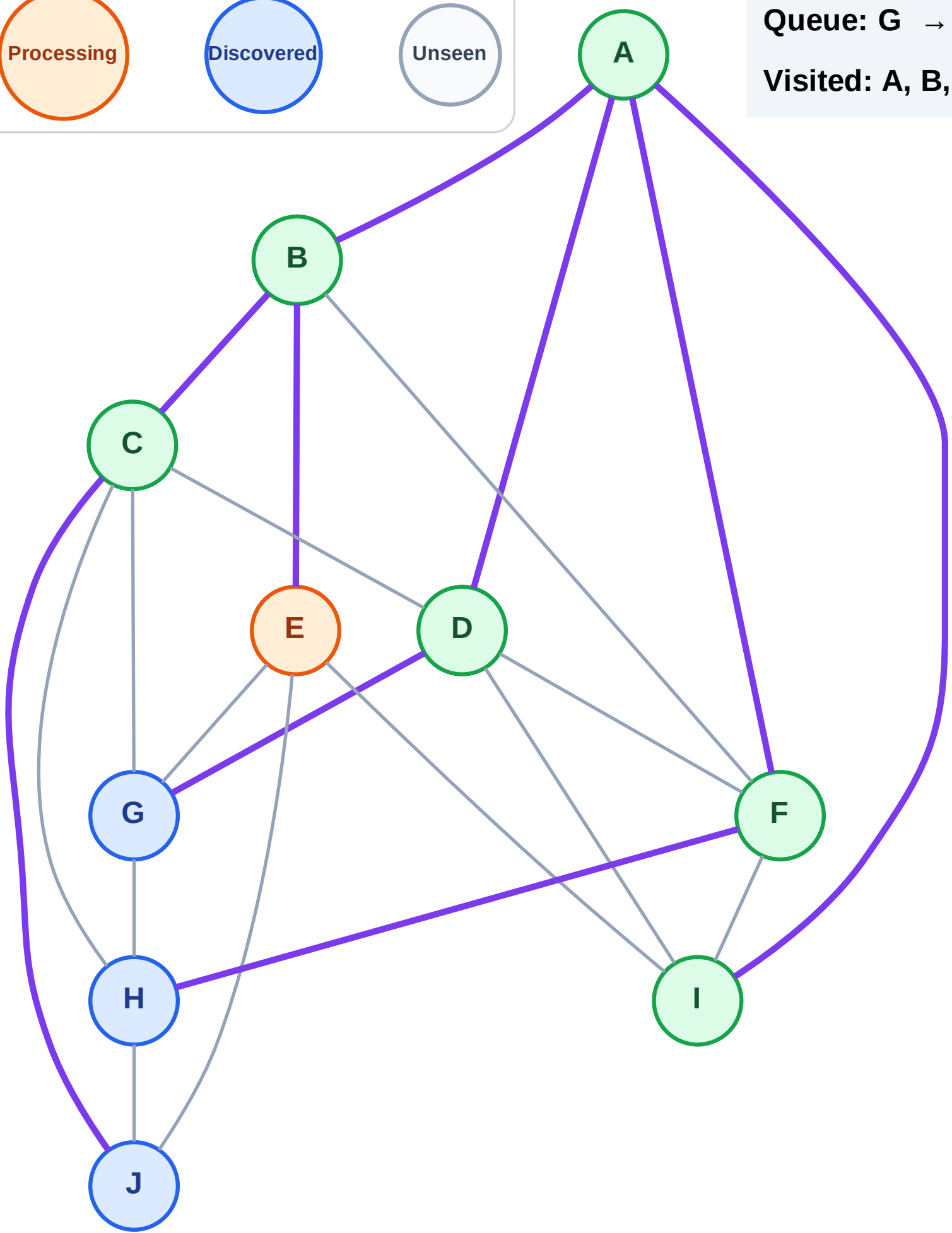
Step 14: Process node E

Legend



Queue: G → H → J

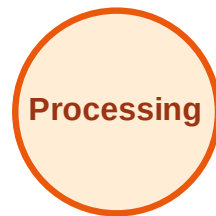
Visited: A, B, C, D, F, I



BFS Traversal

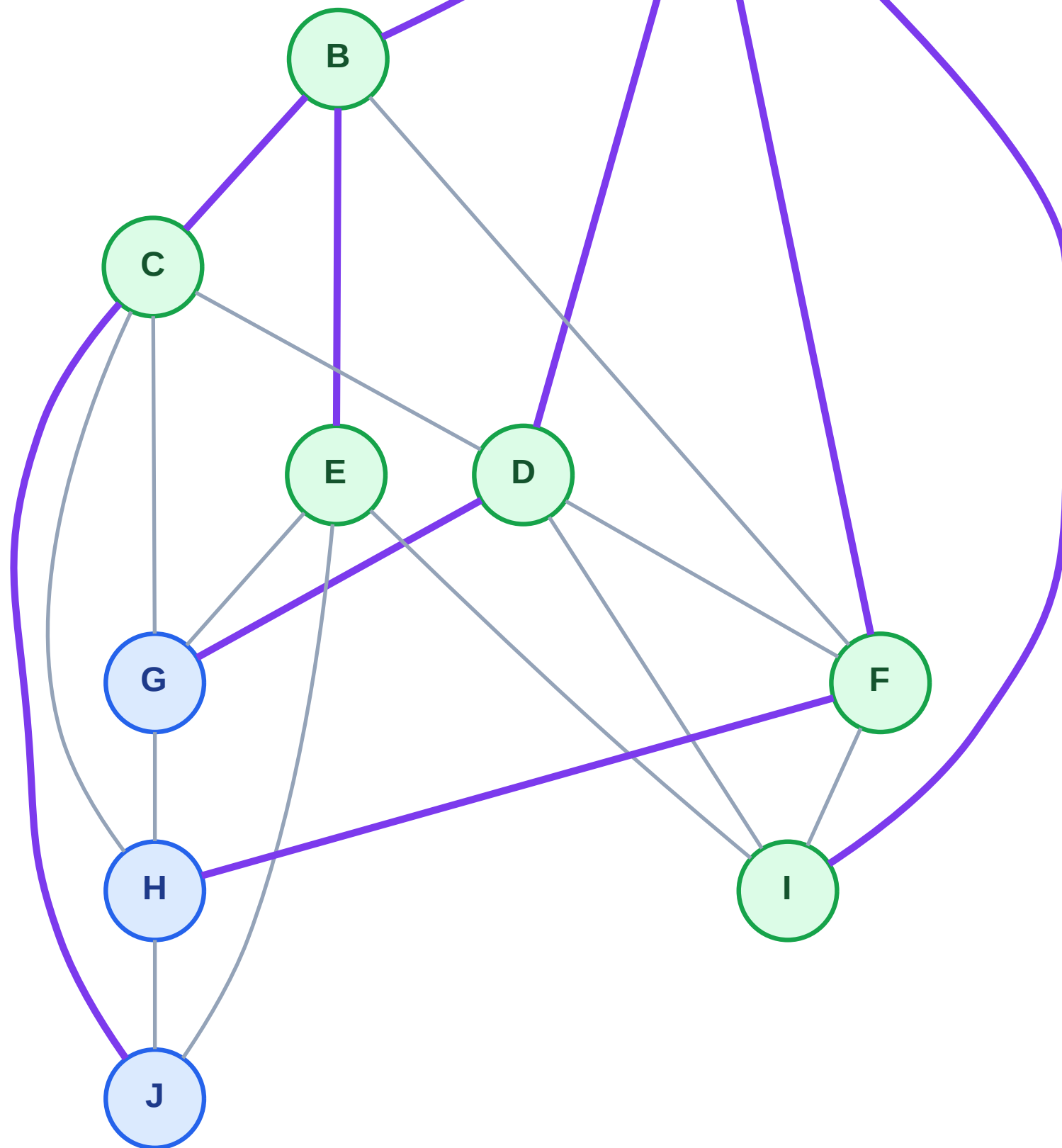
Step 15: No new neighbors from E

Legend



Queue: G → H → J

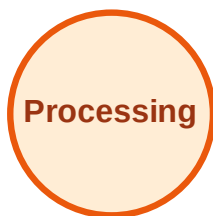
Visited: A, B, C, D, E, F, I



BFS Traversal

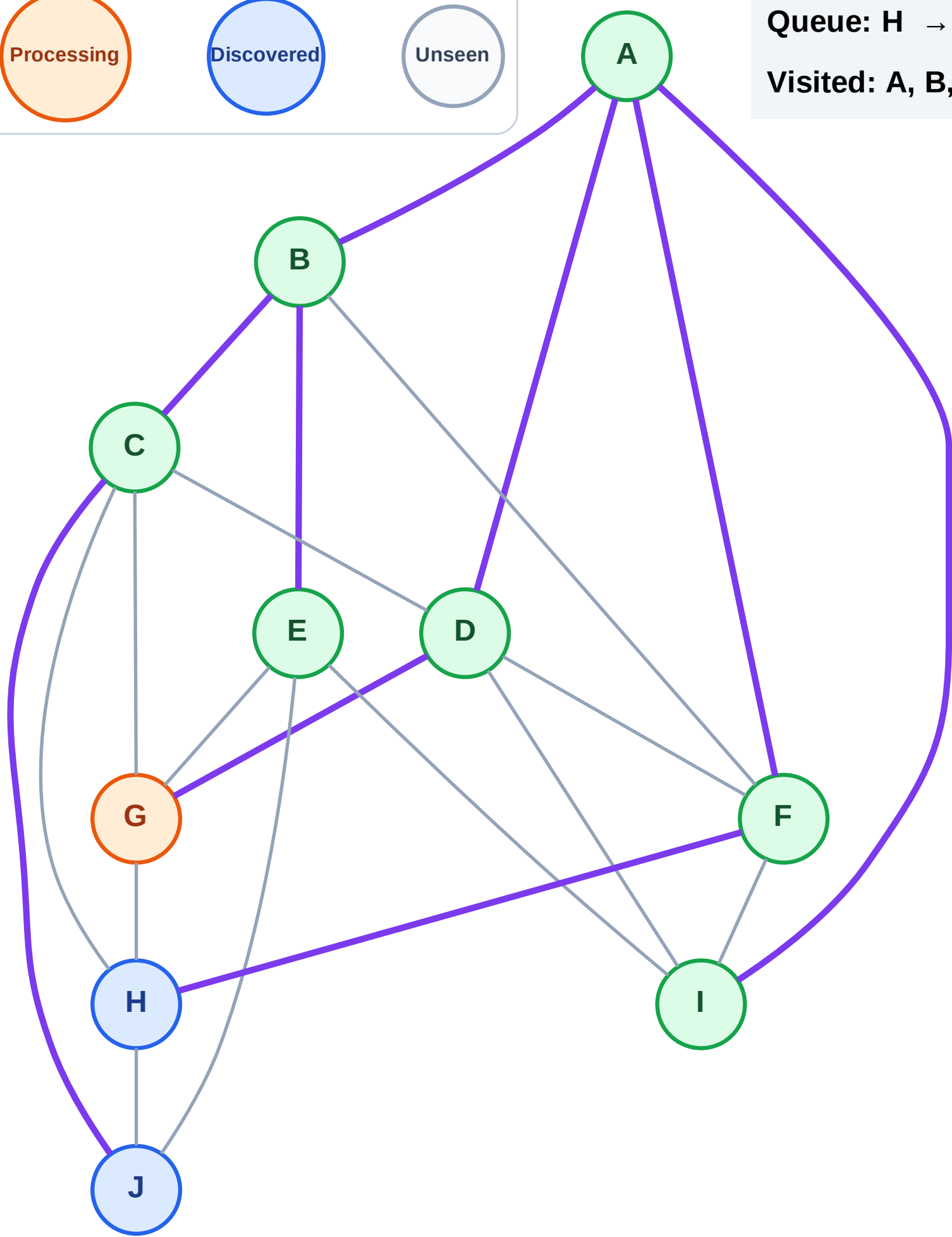
Step 16: Process node G

Legend



Queue: H → J

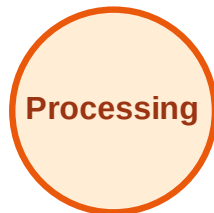
Visited: A, B, C, D, E, F, I



BFS Traversal

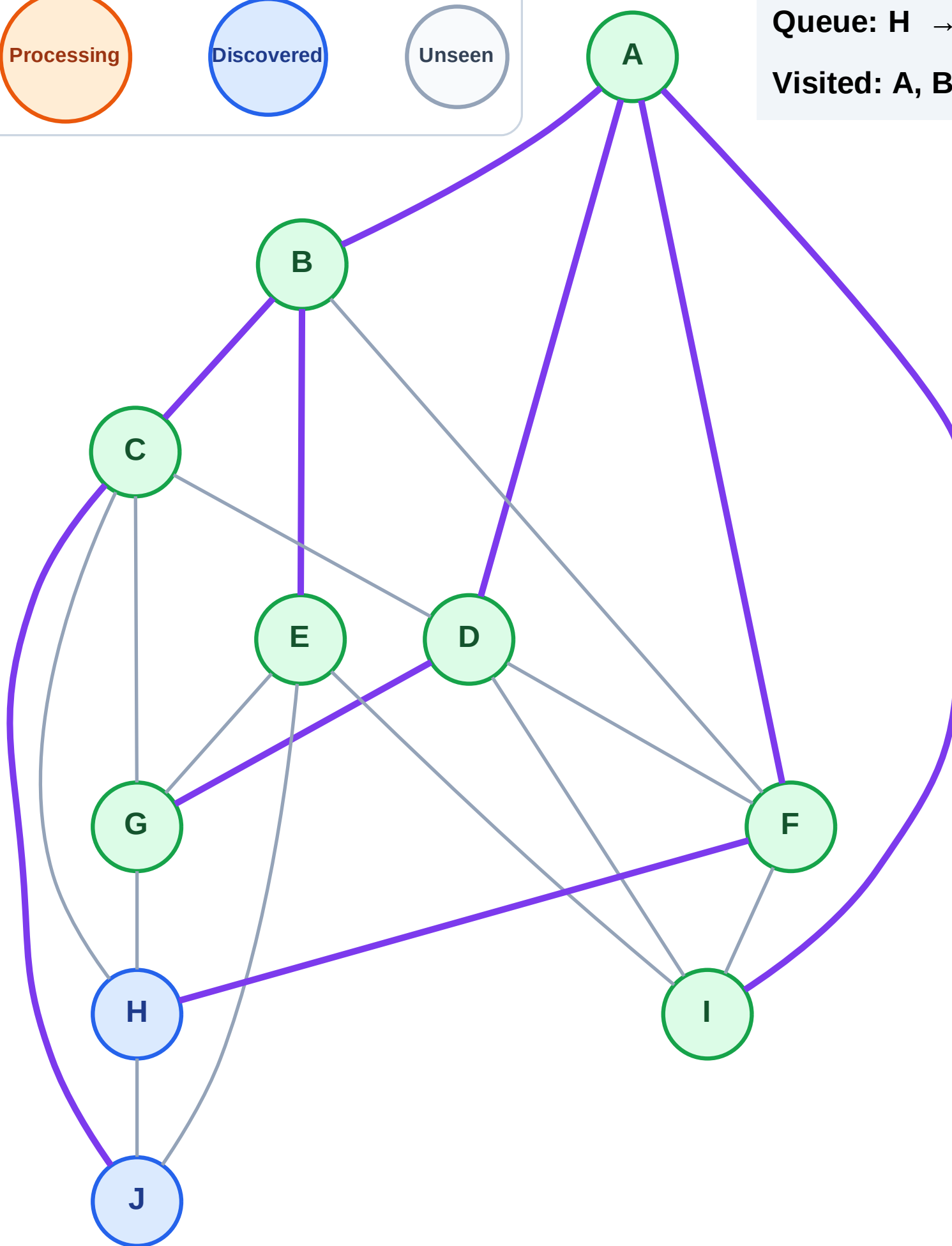
Step 17: No new neighbors from G

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

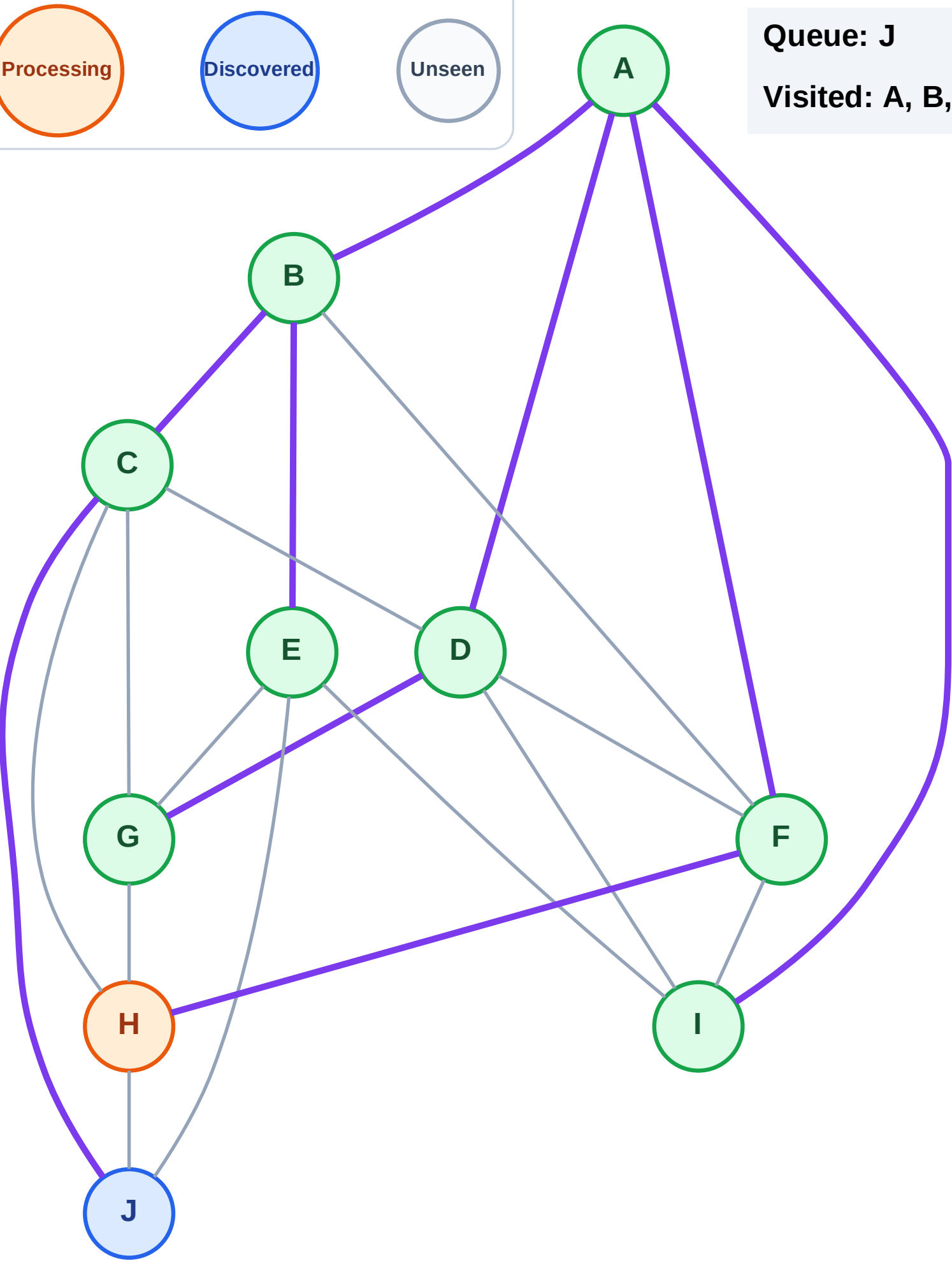
Processing

Discovered

Unseen

Queue: J

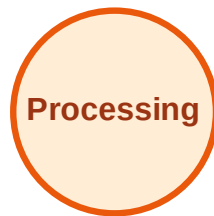
Visited: A, B, C, D, E, F, G, I



BFS Traversal

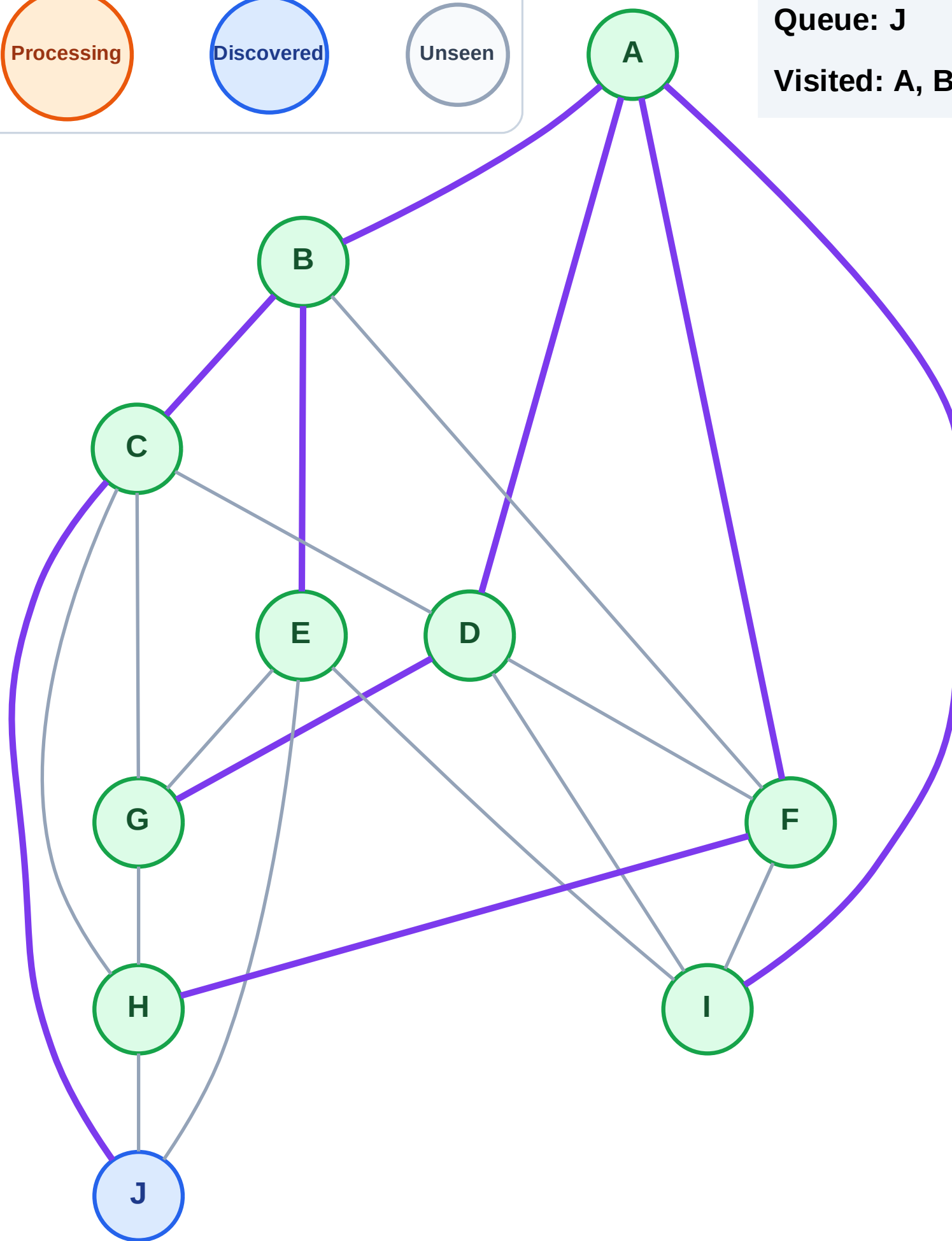
Step 19: No new neighbors from H

Legend



Queue: J

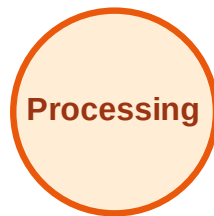
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

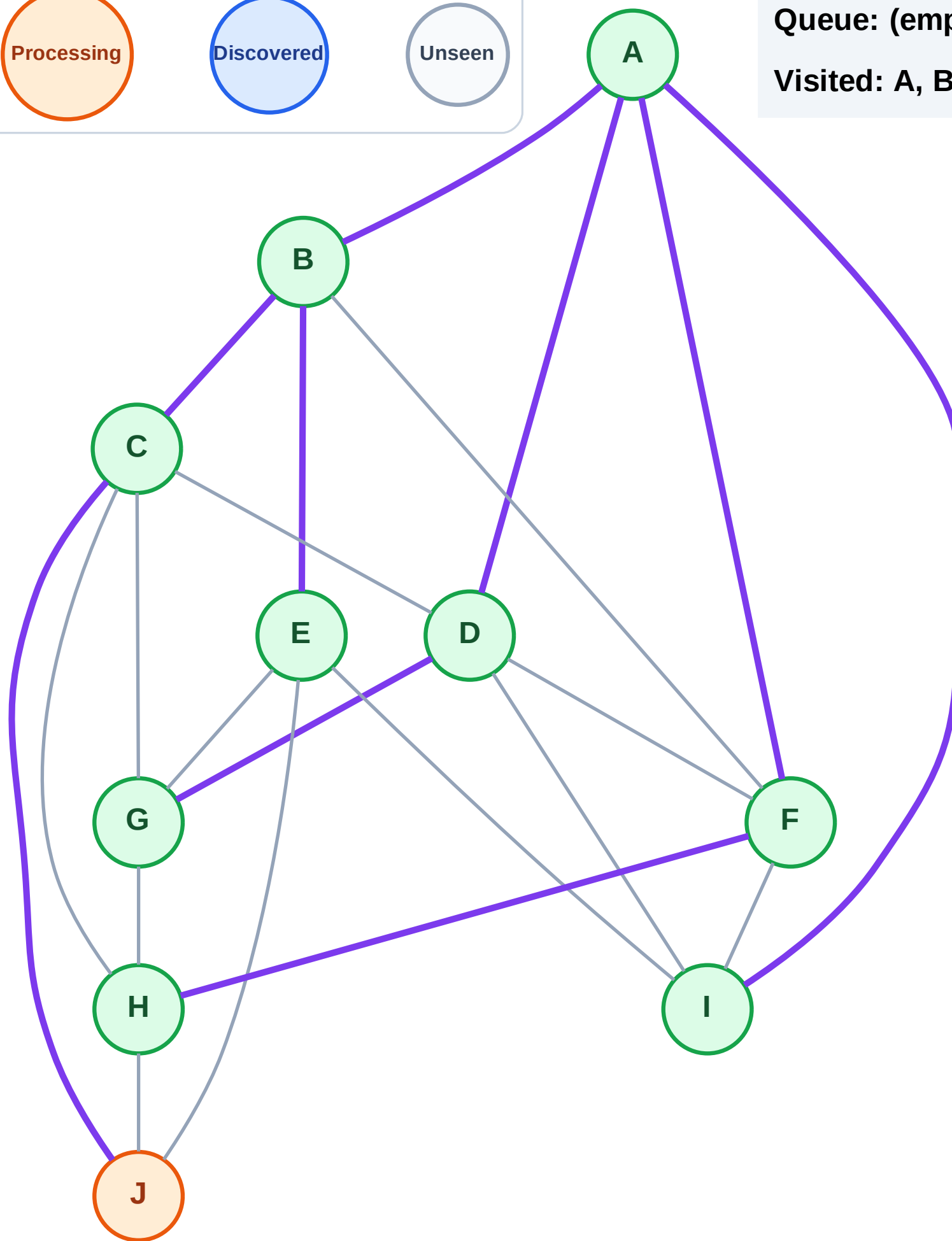
Step 20: Process node J

Legend



Queue: (empty)

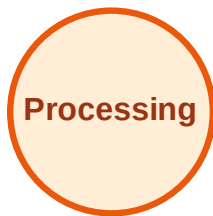
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

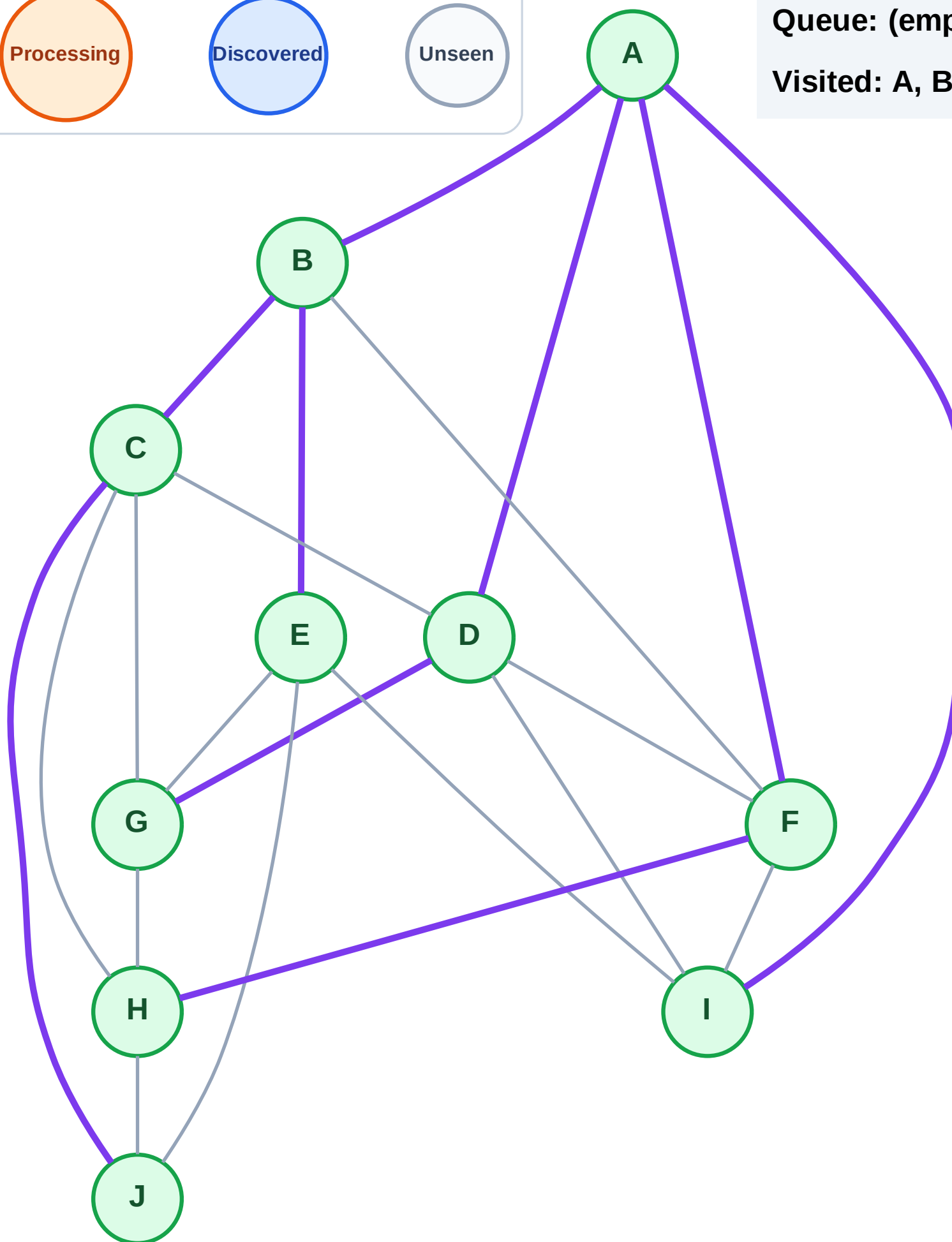
Step 21: No new neighbors from J

Legend



Queue: (empty)

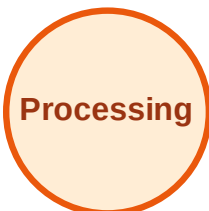
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

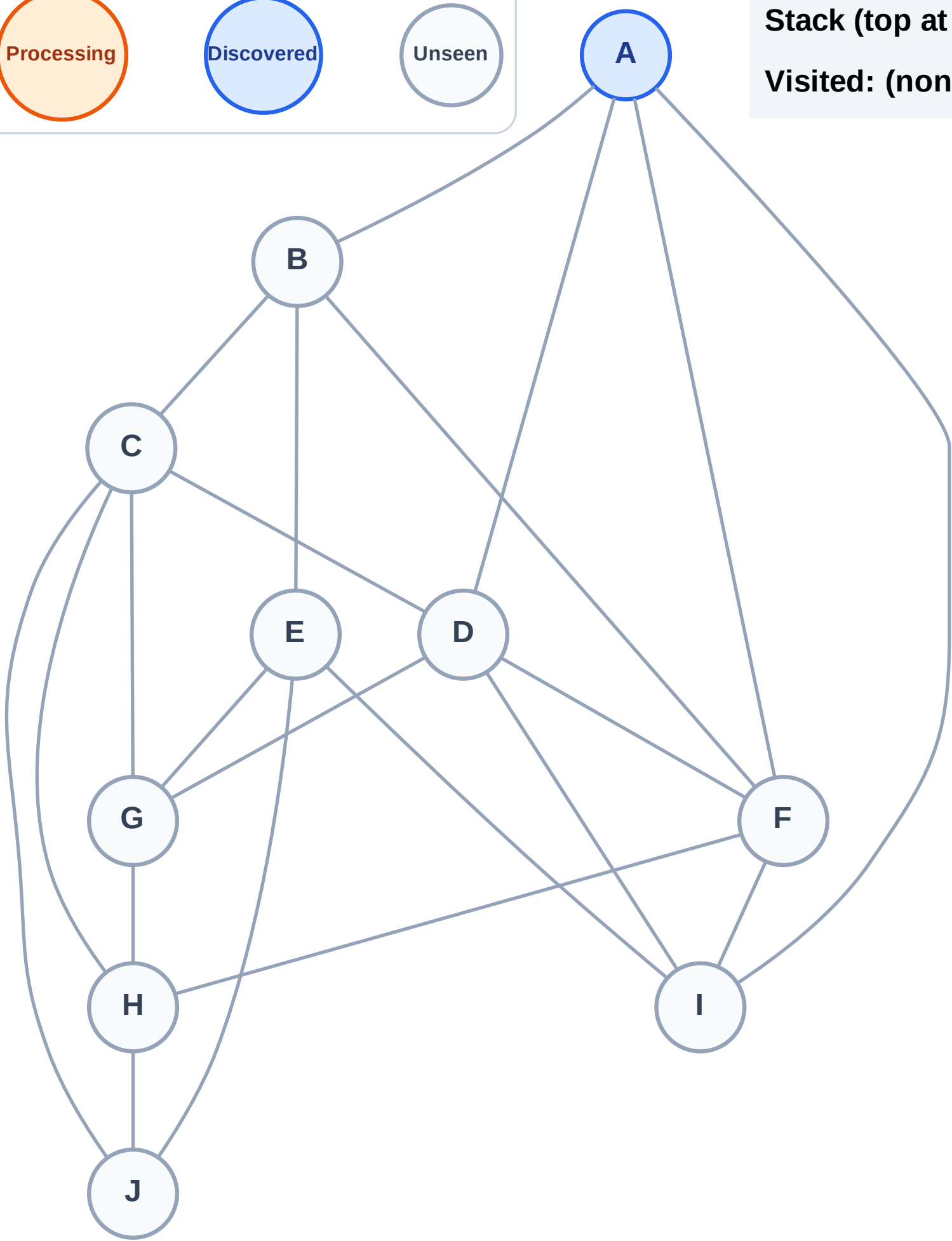
Step 1: Start at A

Legend



Stack (top at right): A

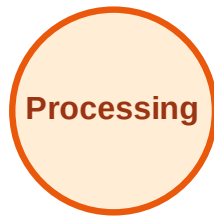
Visited: (none)



DFS Traversal

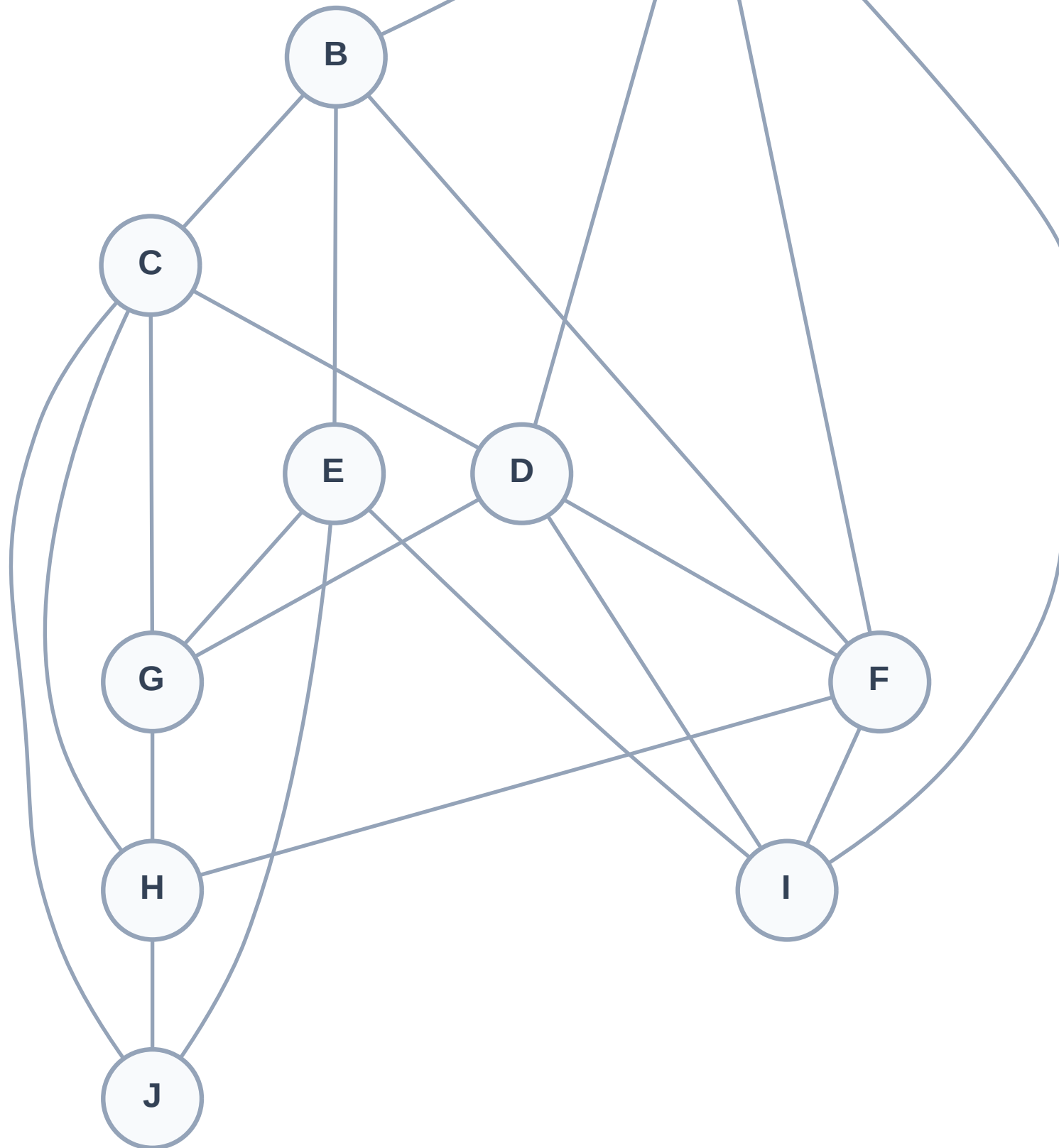
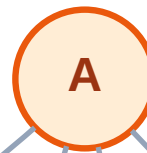
Step 2: Process node A

Legend



Stack (top at right): (empty)

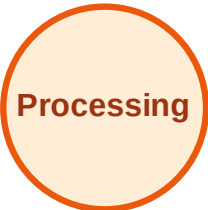
Visited: (none)



DFS Traversal

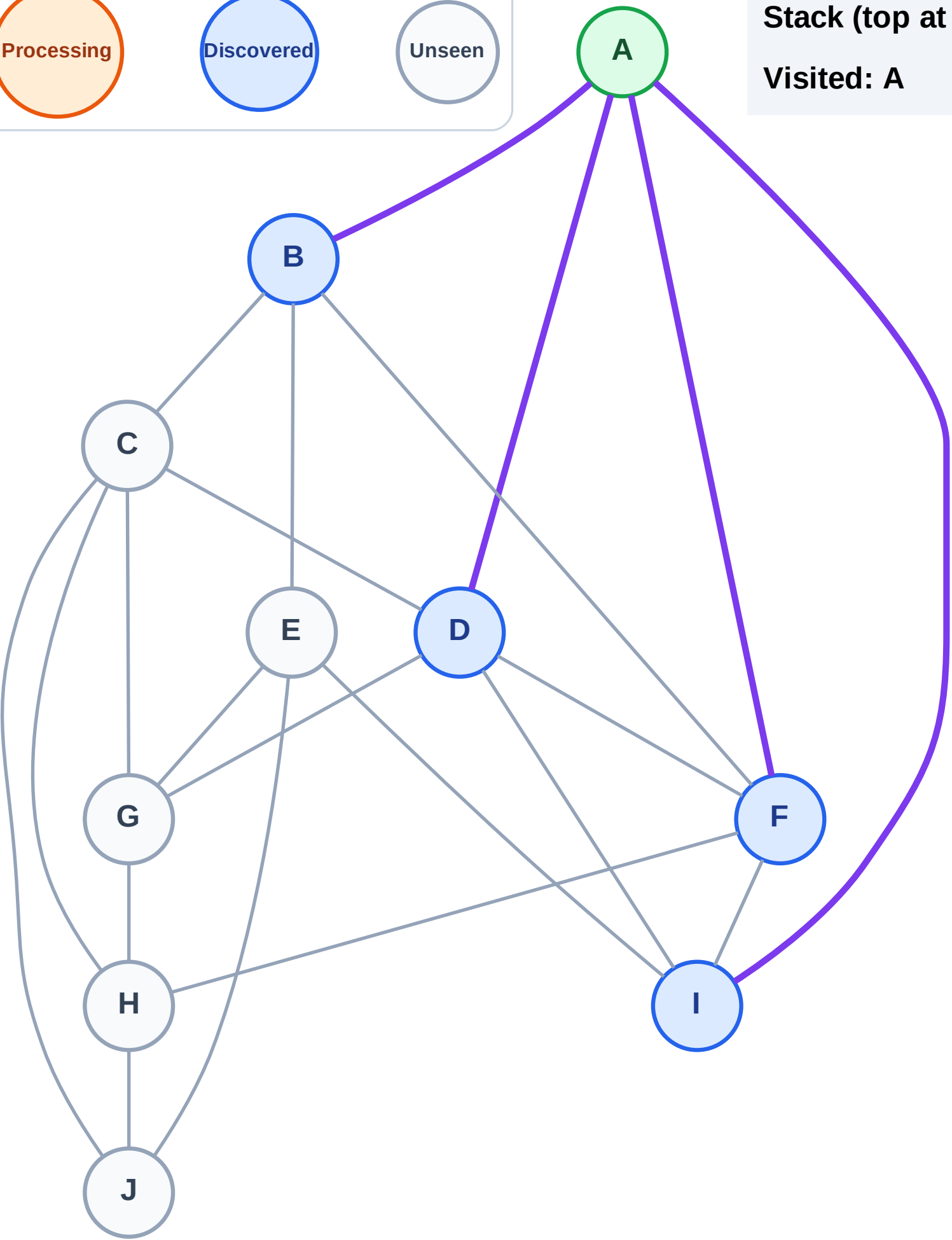
Step 3: Discover I, F, D, B from A

Legend



Stack (top at right): I | F | D | B

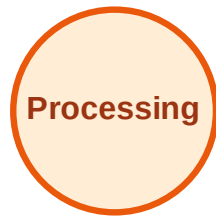
Visited: A



DFS Traversal

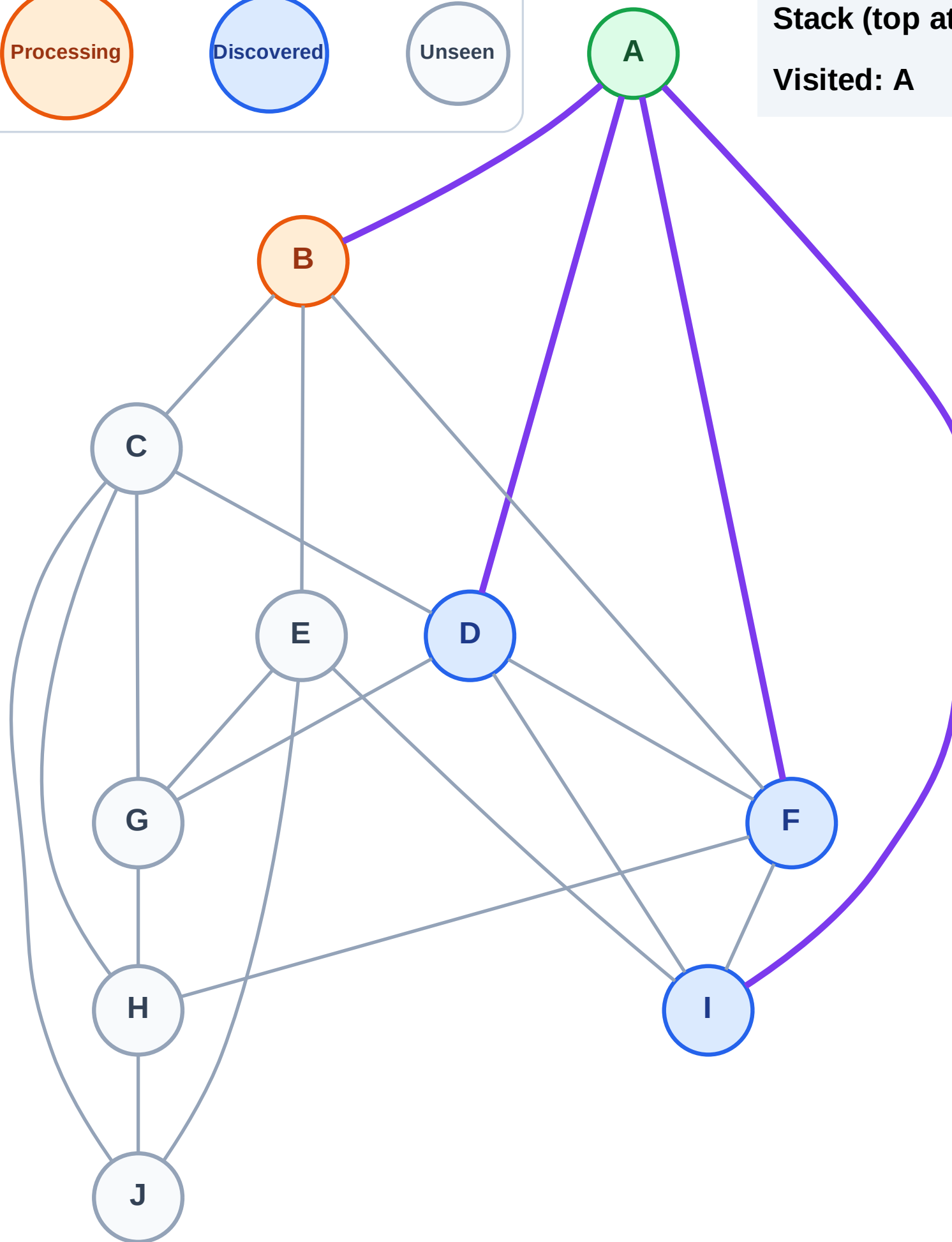
Step 4: Process node B

Legend



Stack (top at right): I | F | D

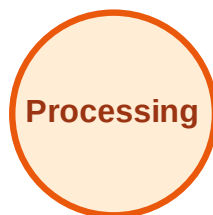
Visited: A



DFS Traversal

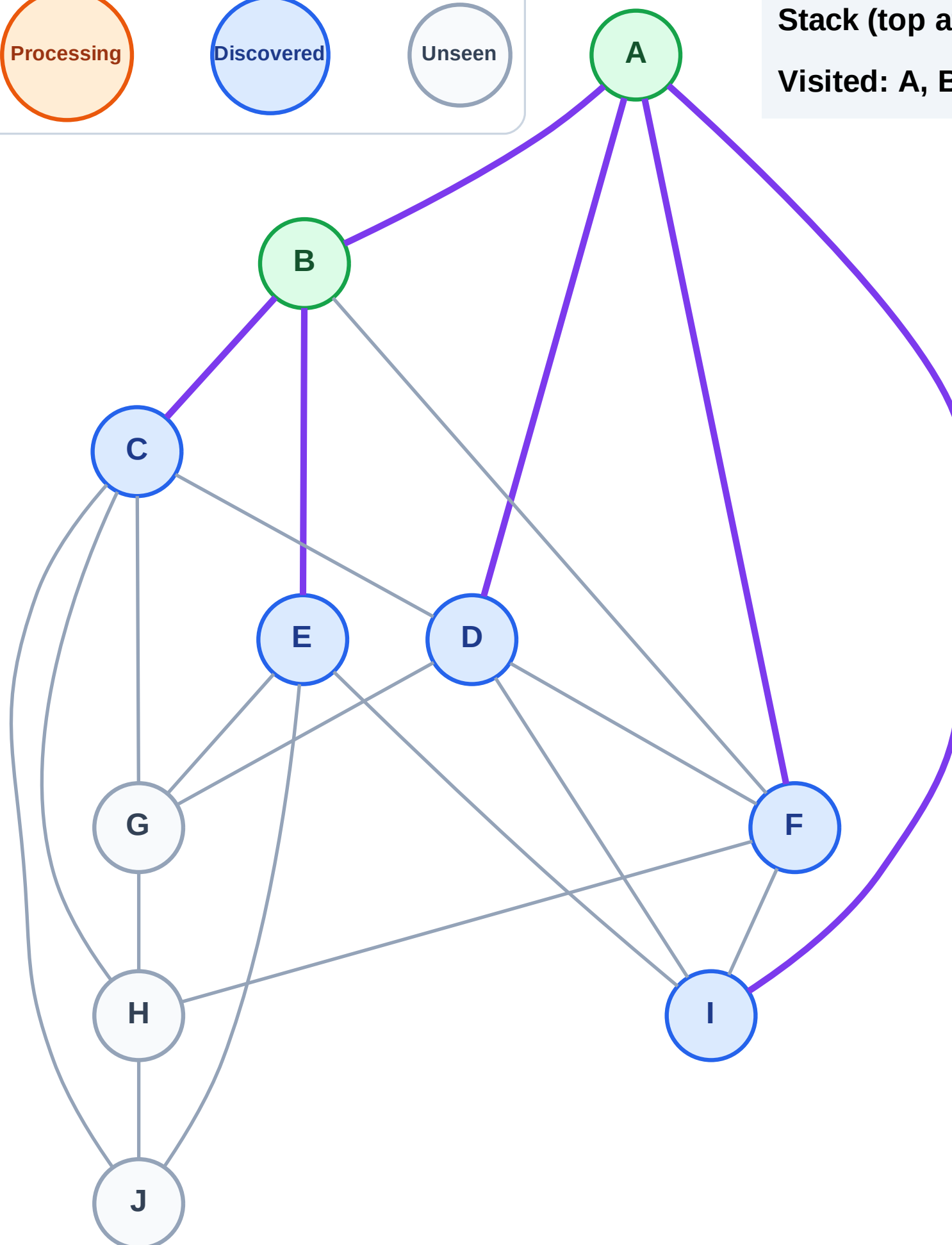
Step 5: Discover E, C from B

Legend



Stack (top at right): I | F | D | E | C

Visited: A, B



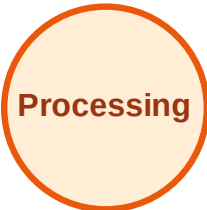
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



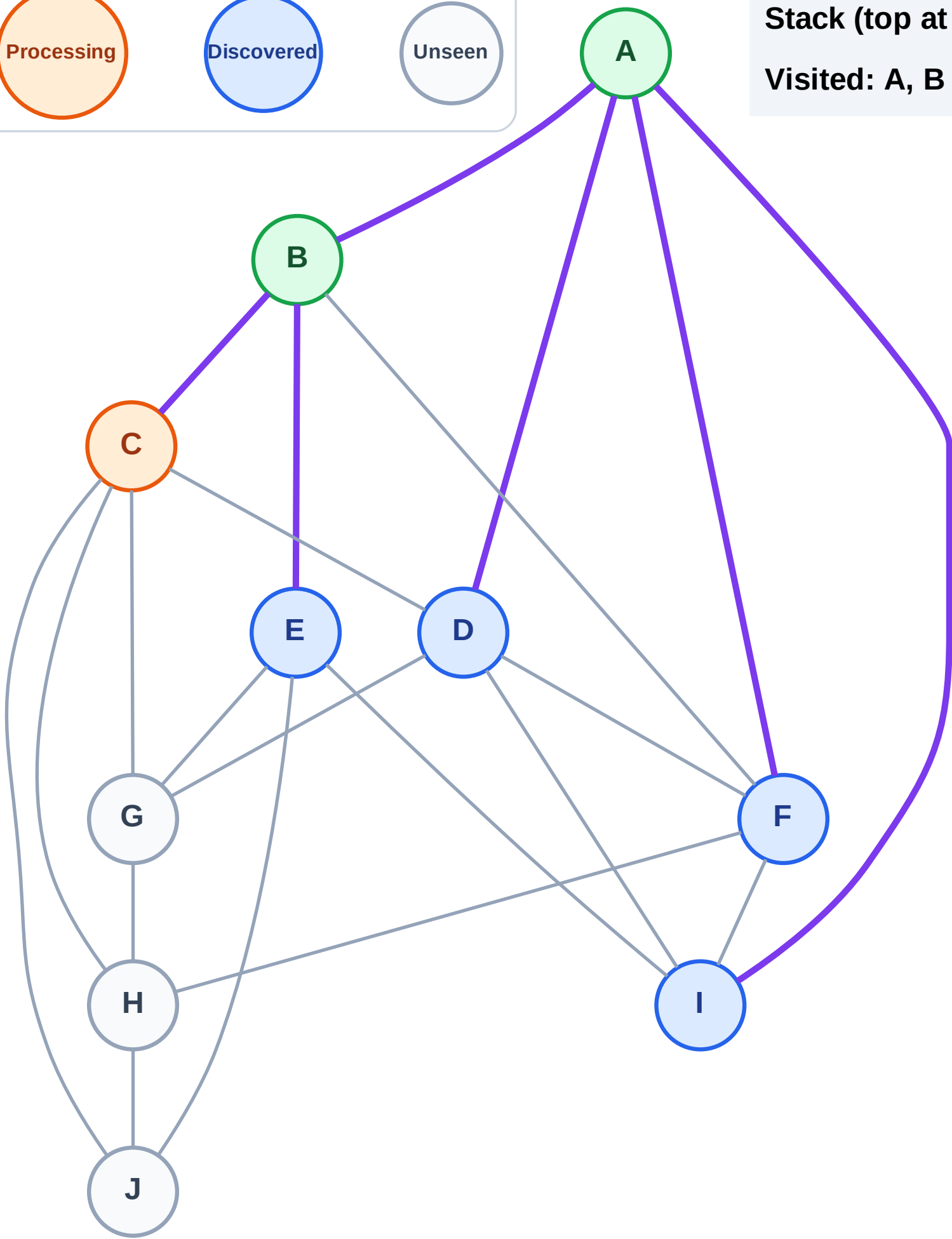
Discovered



Unseen

Stack (top at right): I | F | D | E

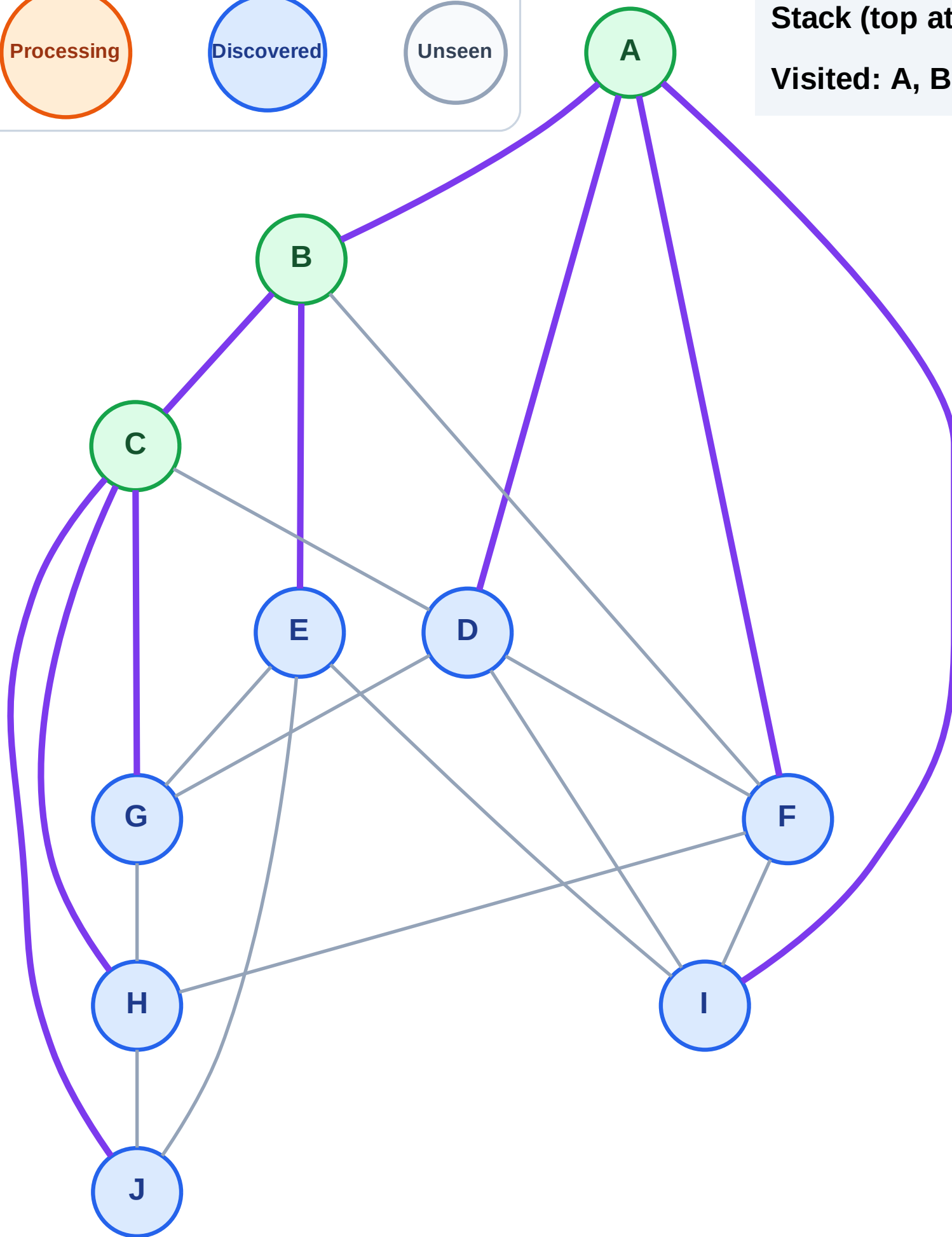
Visited: A, B



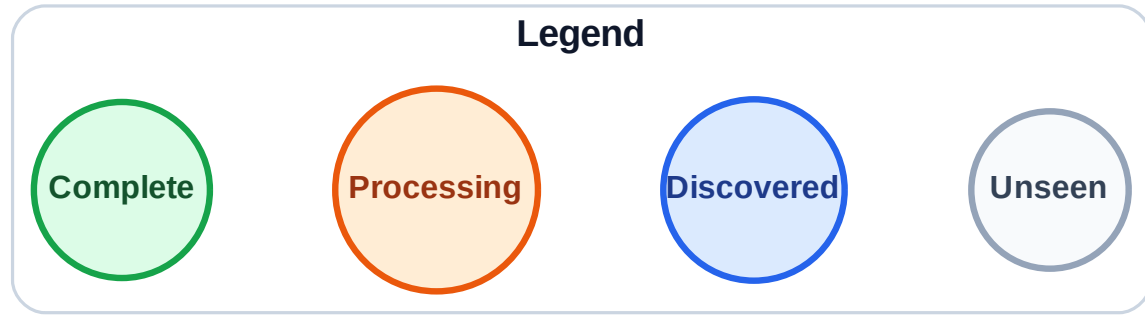
Step 7: Discover J, H, G from C

Stack (top at right): I | F | D | E | J | H | G

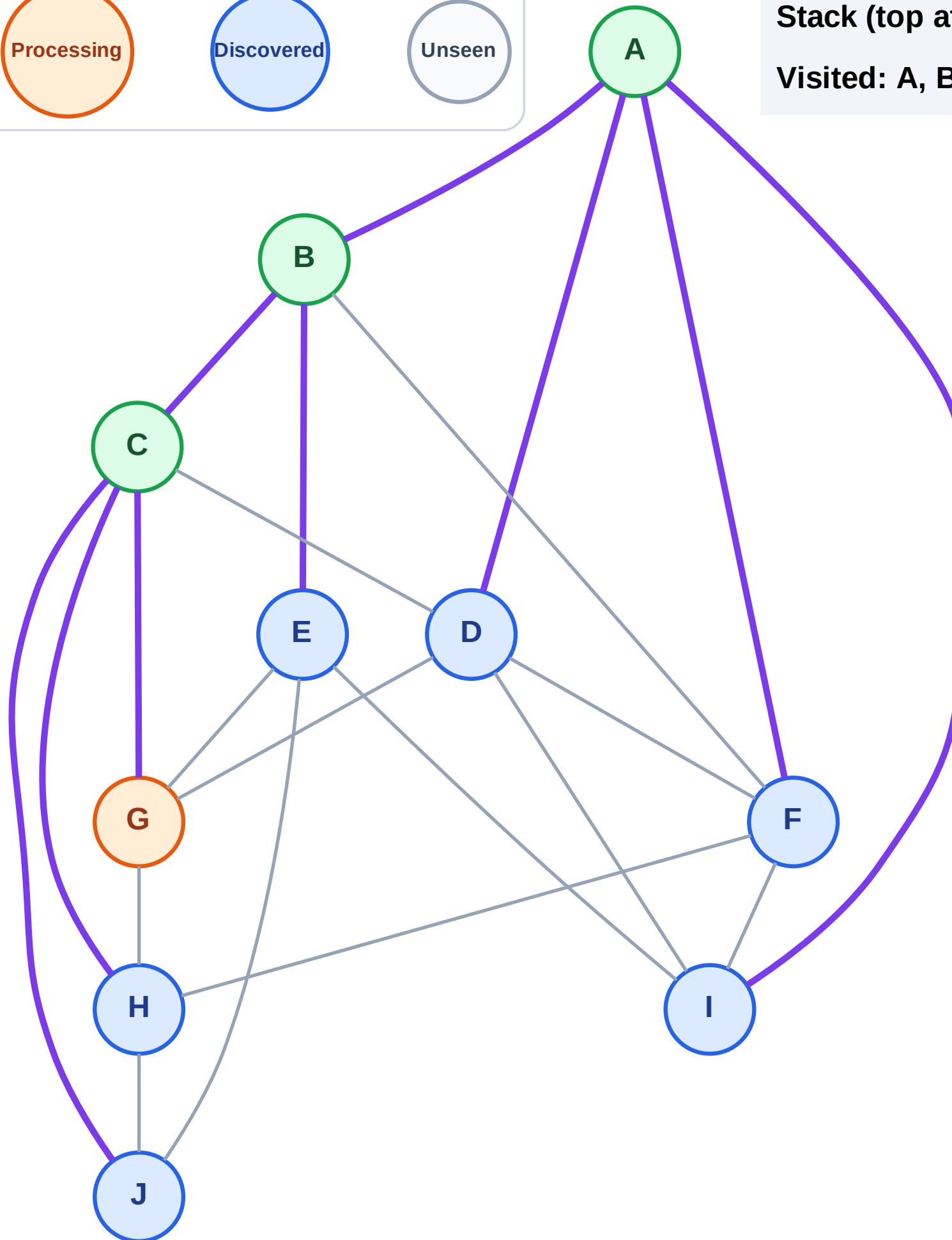
Visited: A, B, C



Step 8: Process node G



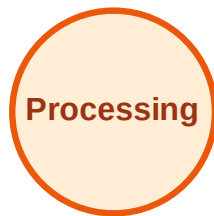
Visited: A, B, C



DFS Traversal

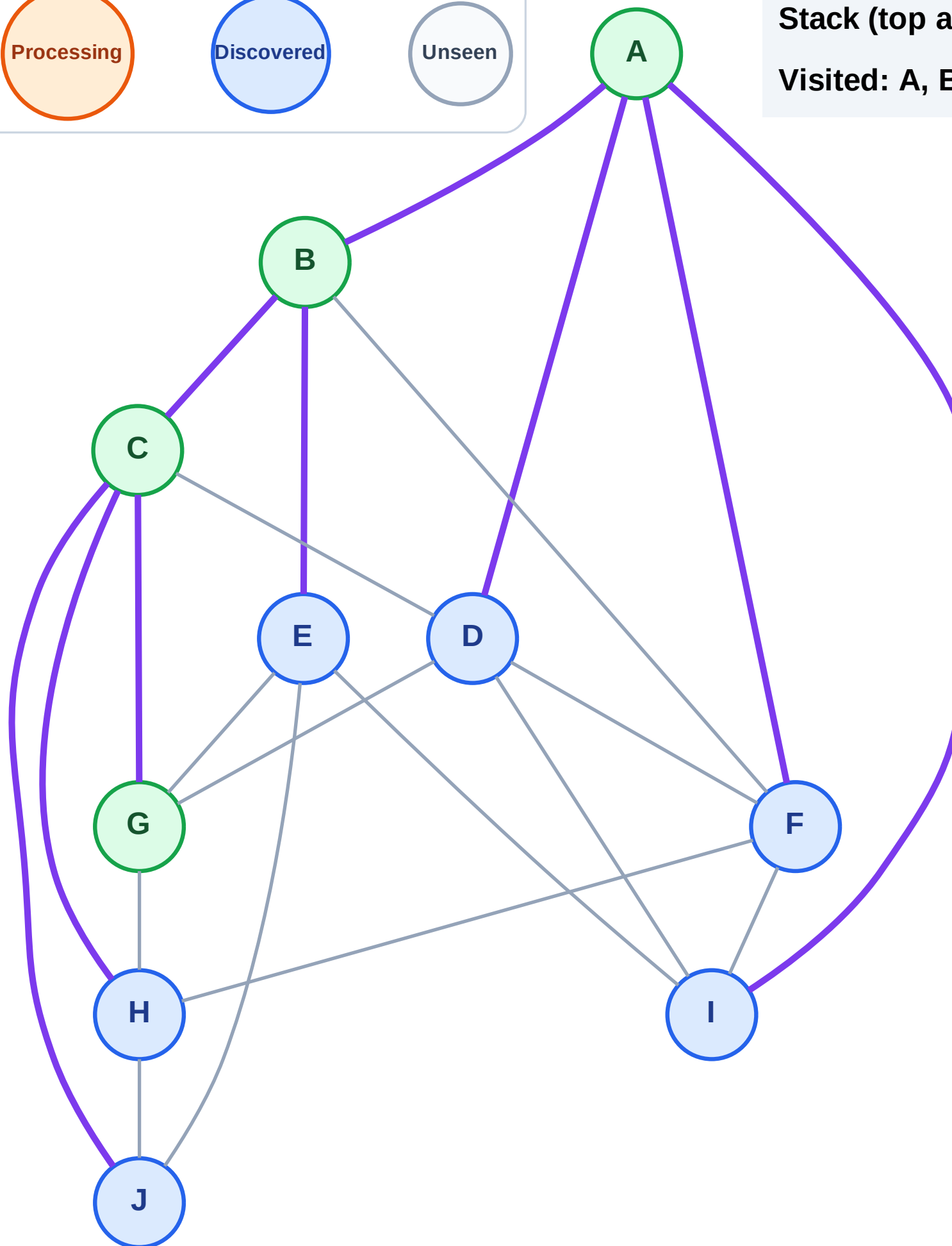
Step 9: No new neighbors from G

Legend



Stack (top at right): I | F | D | E | J | H

Visited: A, B, C, G



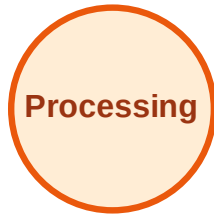
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



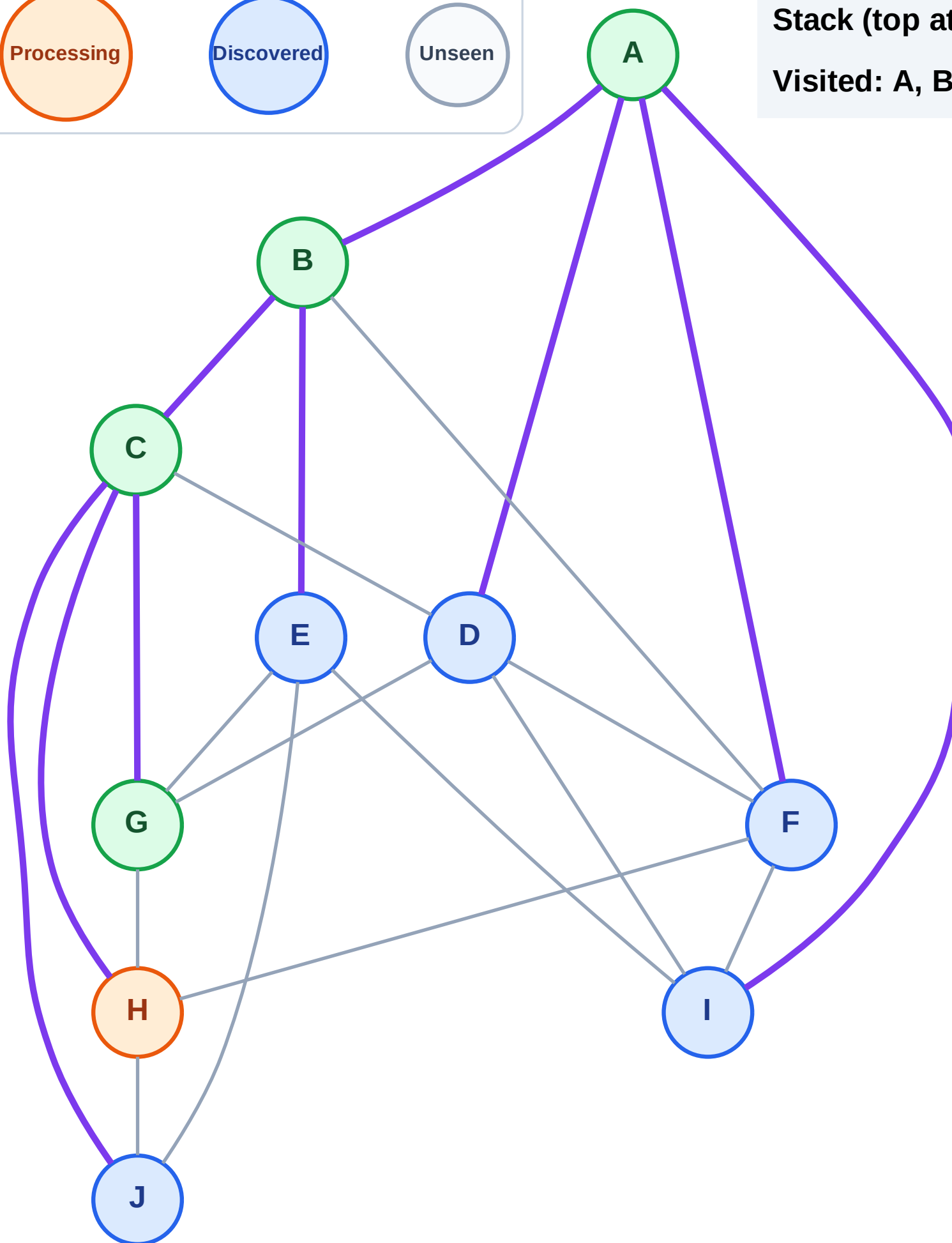
Discovered



Unseen

Stack (top at right): I | F | D | E | J

Visited: A, B, C, G



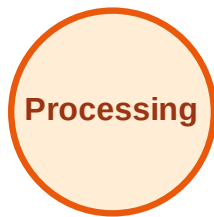
DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



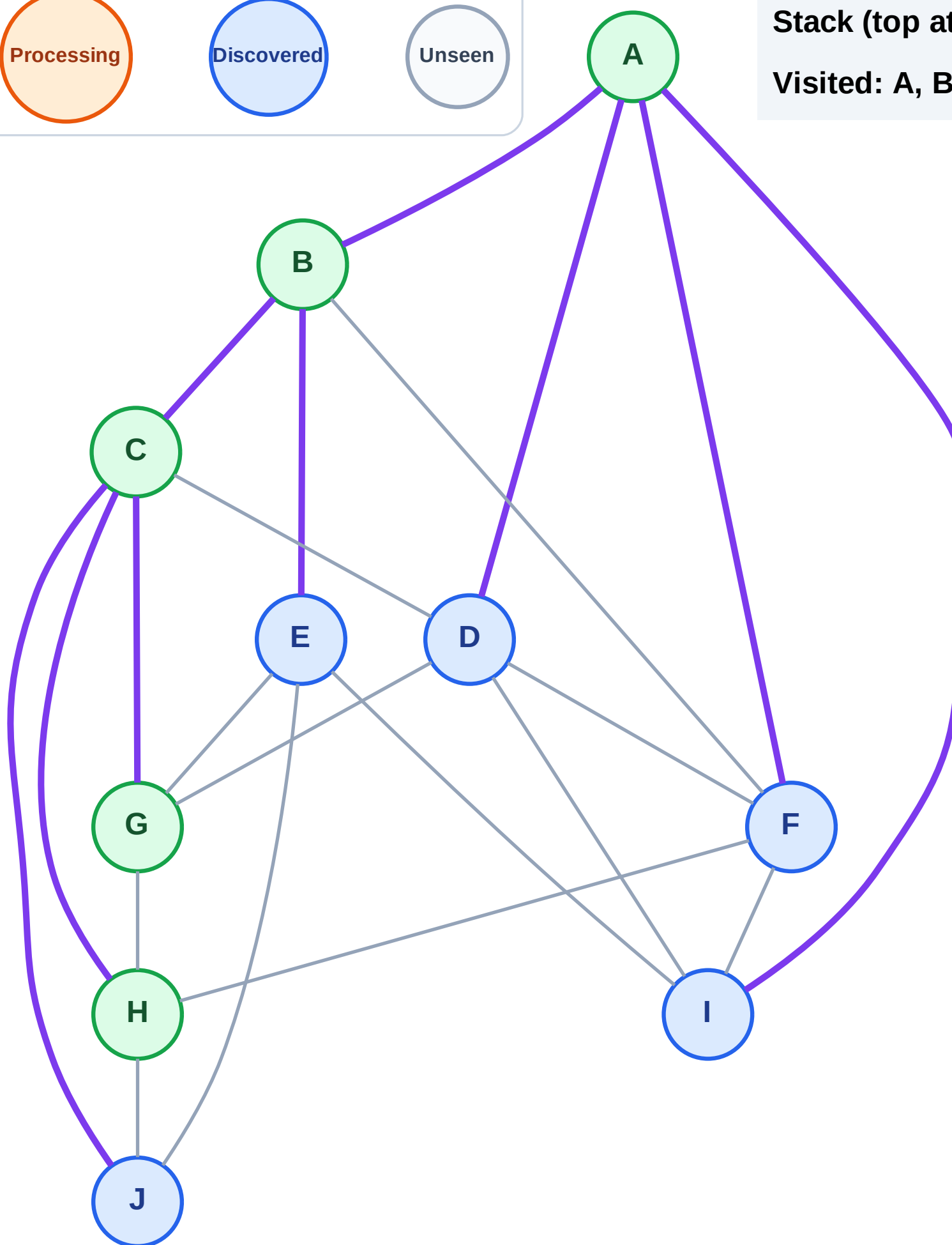
Discovered



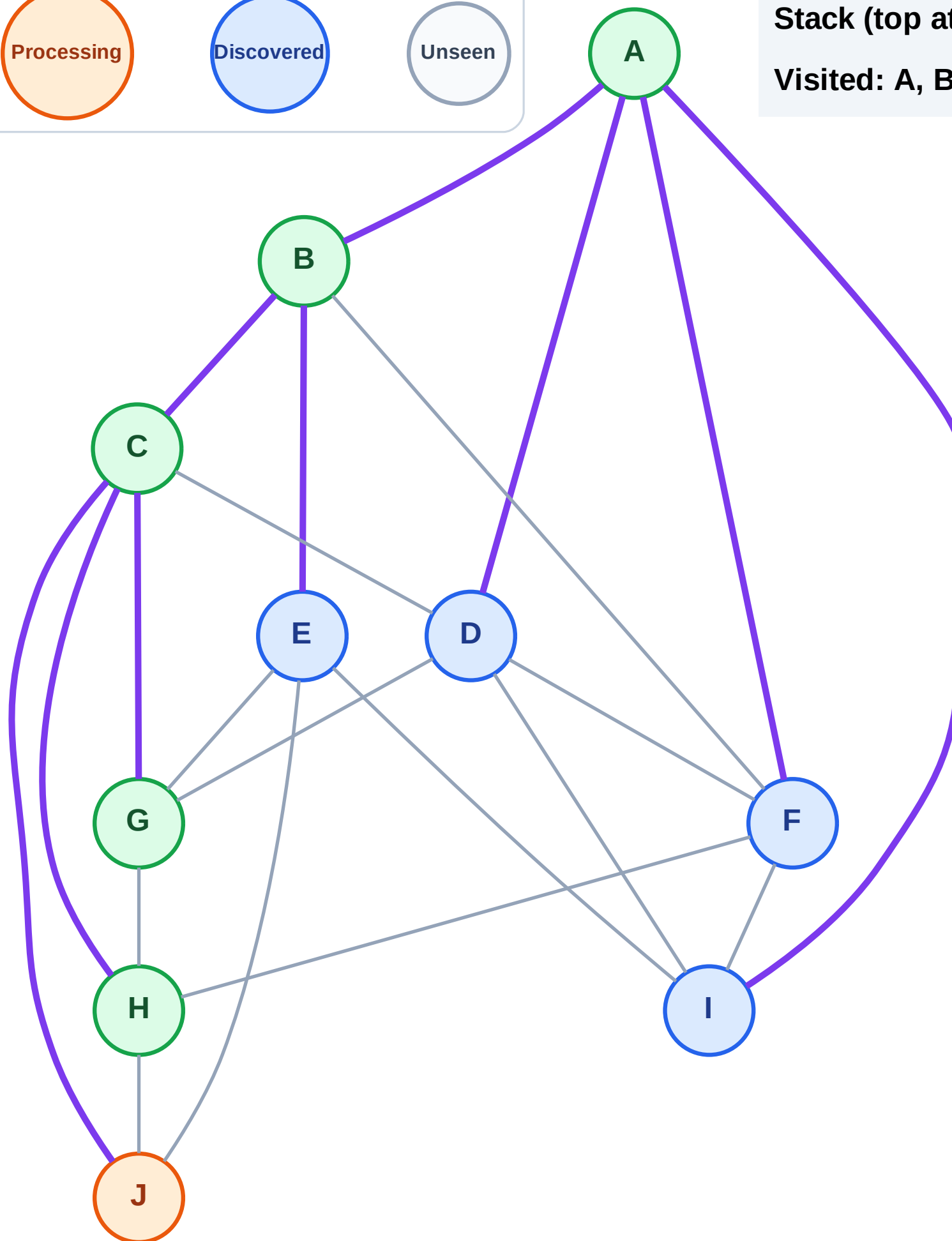
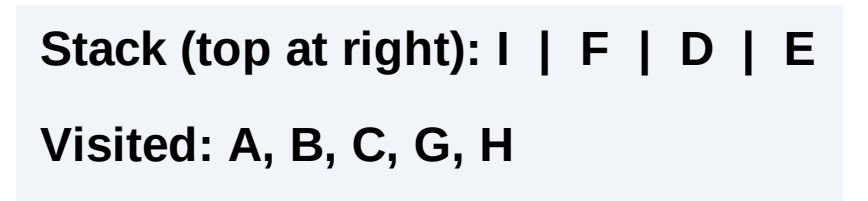
Unseen

Stack (top at right): I | F | D | E | J

Visited: A, B, C, G, H



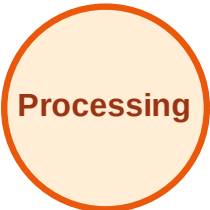
Step 12: Process node J



DFS Traversal

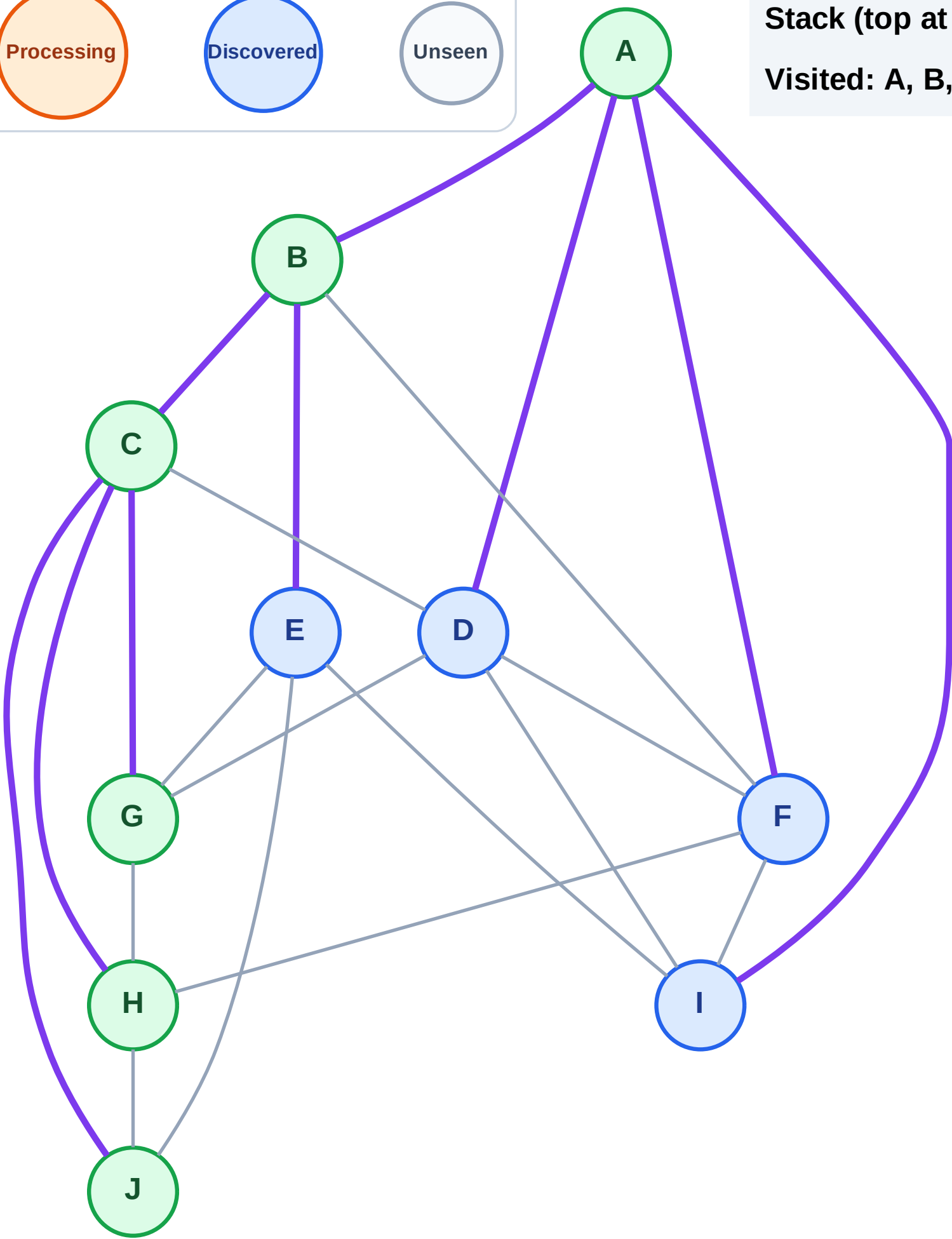
Step 13: No new neighbors from J

Legend



Stack (top at right): I | F | D | E

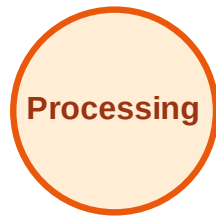
Visited: A, B, C, G, H, J



DFS Traversal

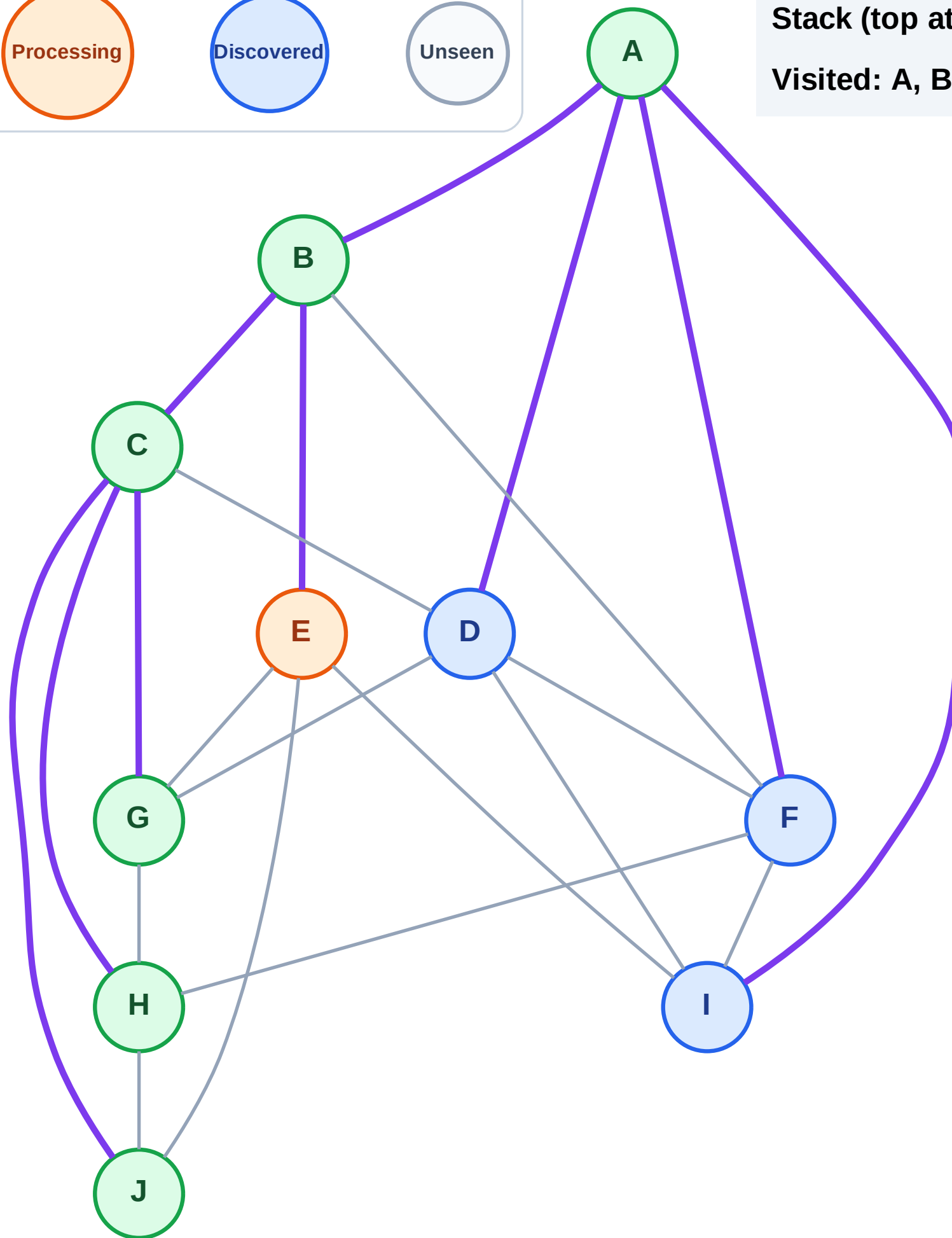
Step 14: Process node E

Legend



Stack (top at right): I | F | D

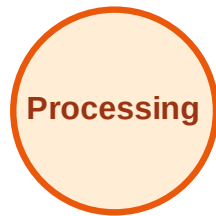
Visited: A, B, C, G, H, J



DFS Traversal

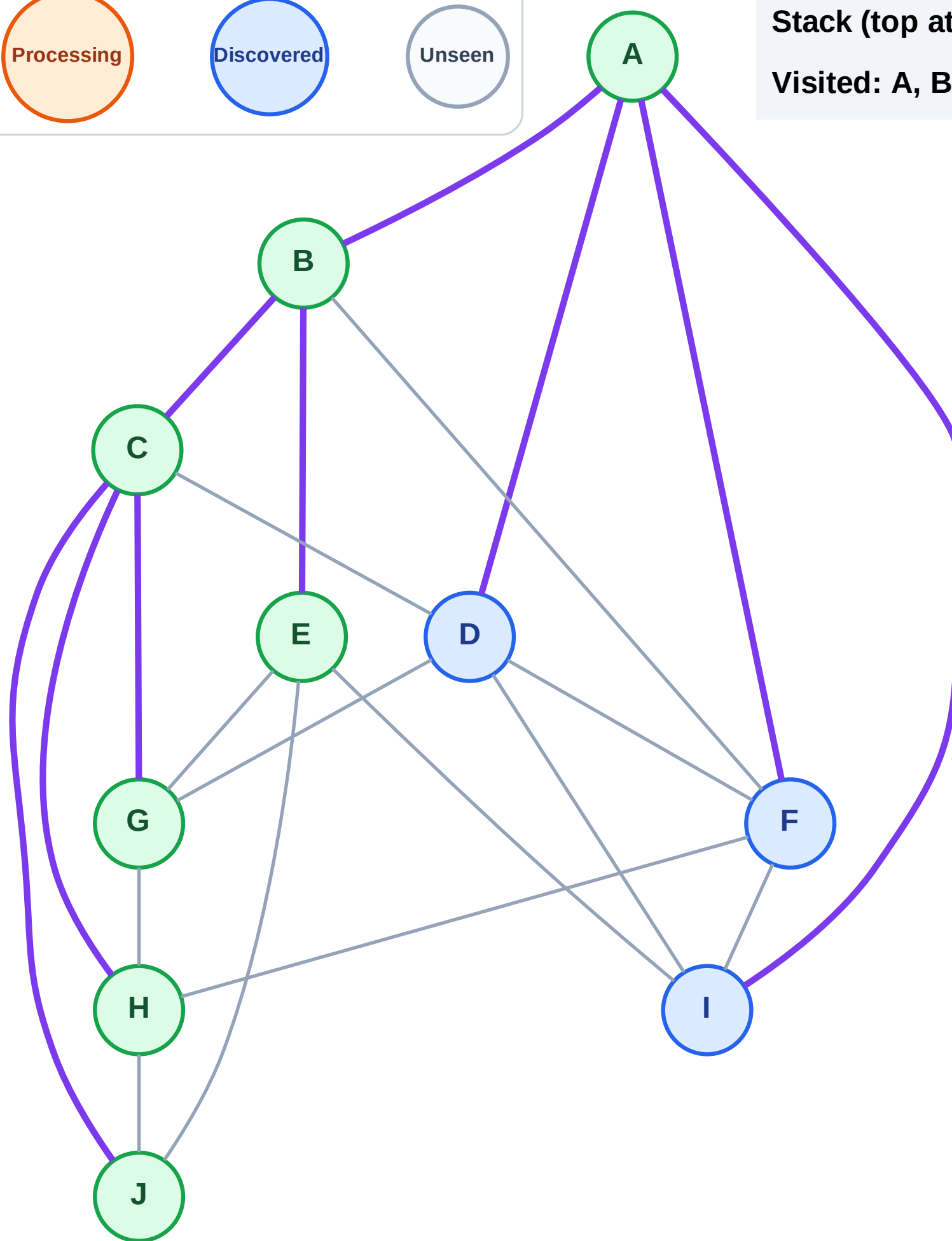
Step 15: No new neighbors from E

Legend



Stack (top at right): I | F | D

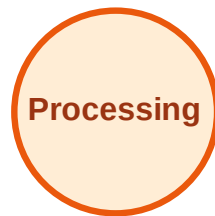
Visited: A, B, C, E, G, H, J



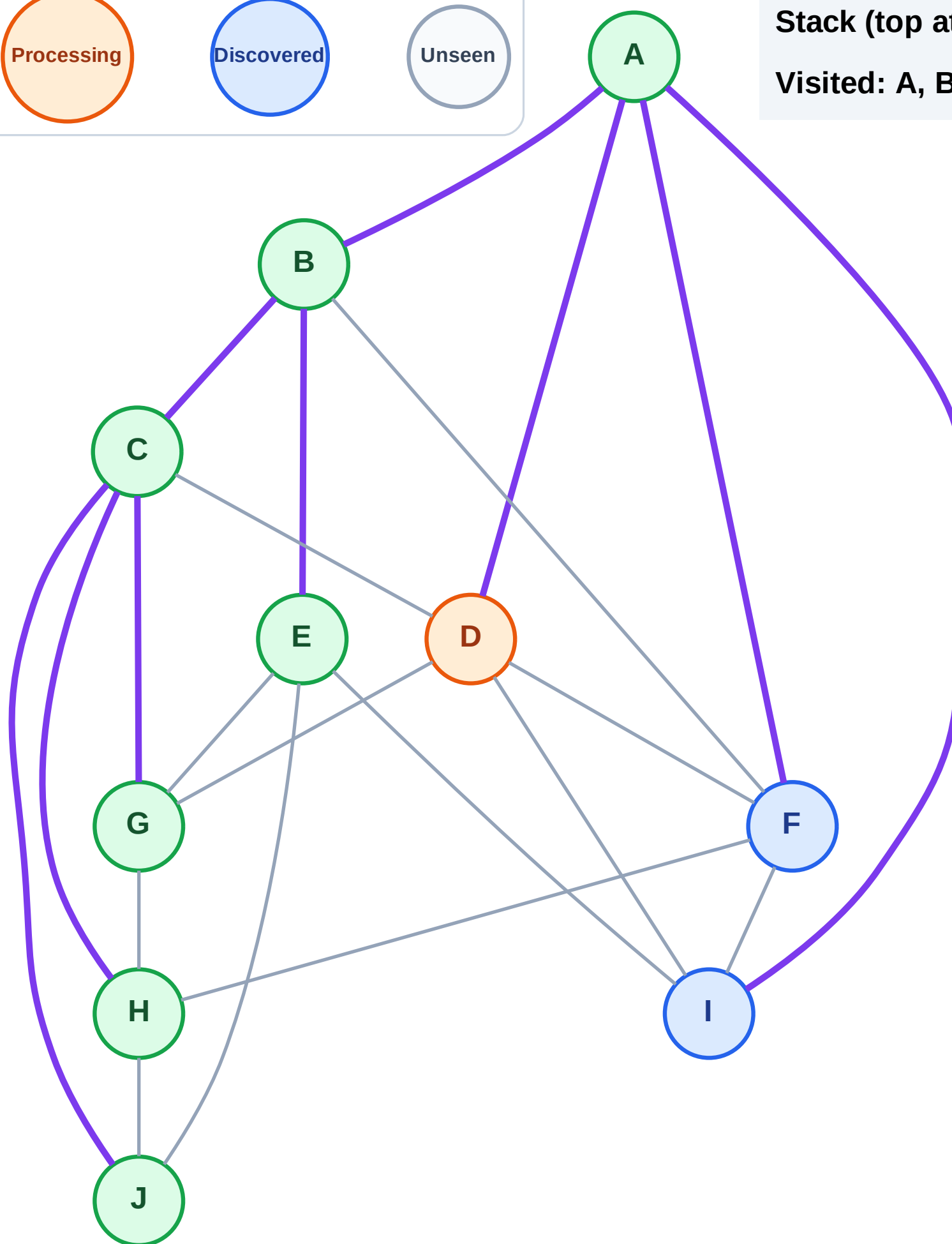
DFS Traversal

Step 16: Process node D

Legend



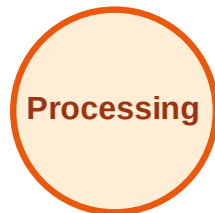
Stack (top at right): I | F
Visited: A, B, C, E, G, H, J



DFS Traversal

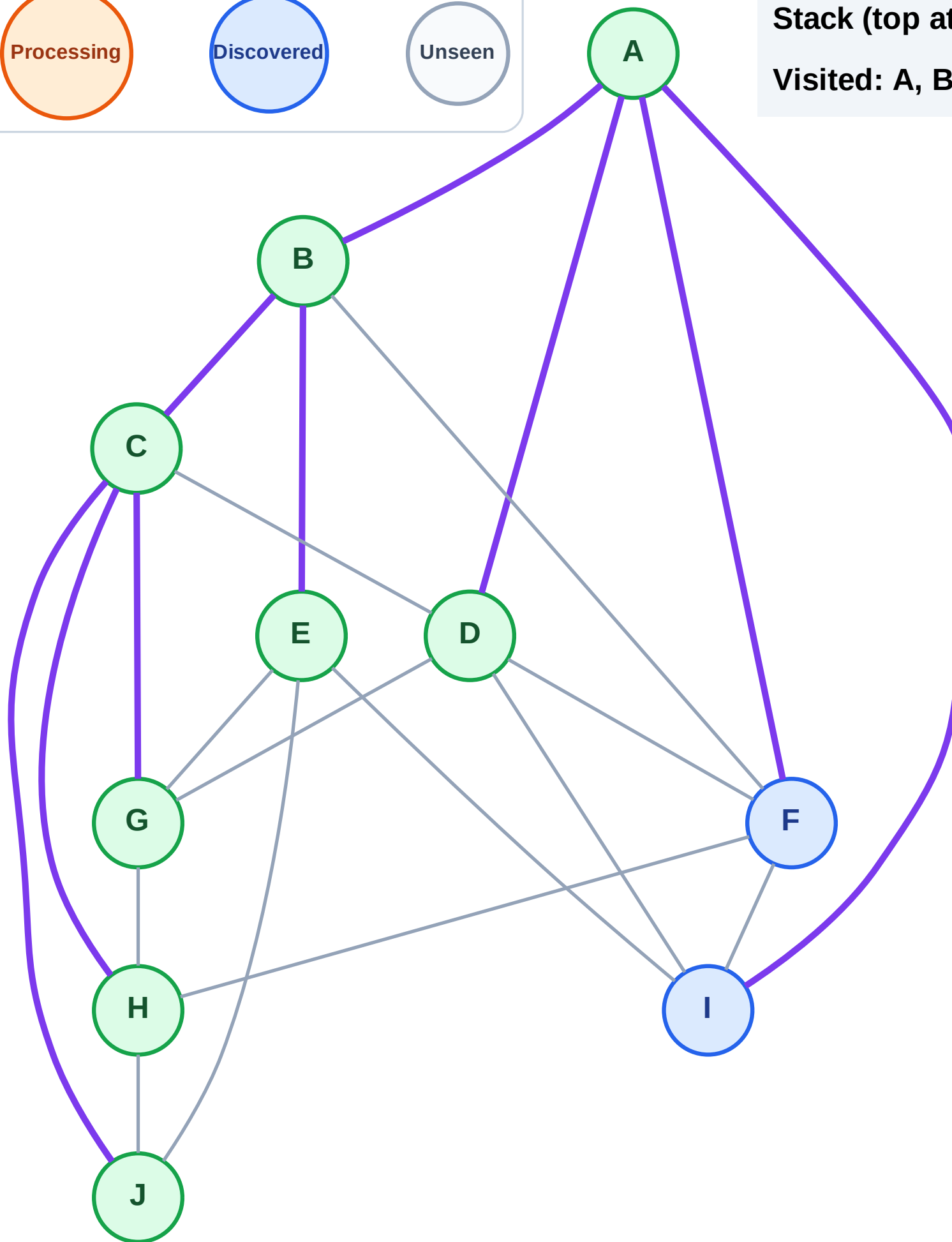
Step 17: No new neighbors from D

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, G, H, J



DFS Traversal

Step 18: Process node F

Legend

Complete

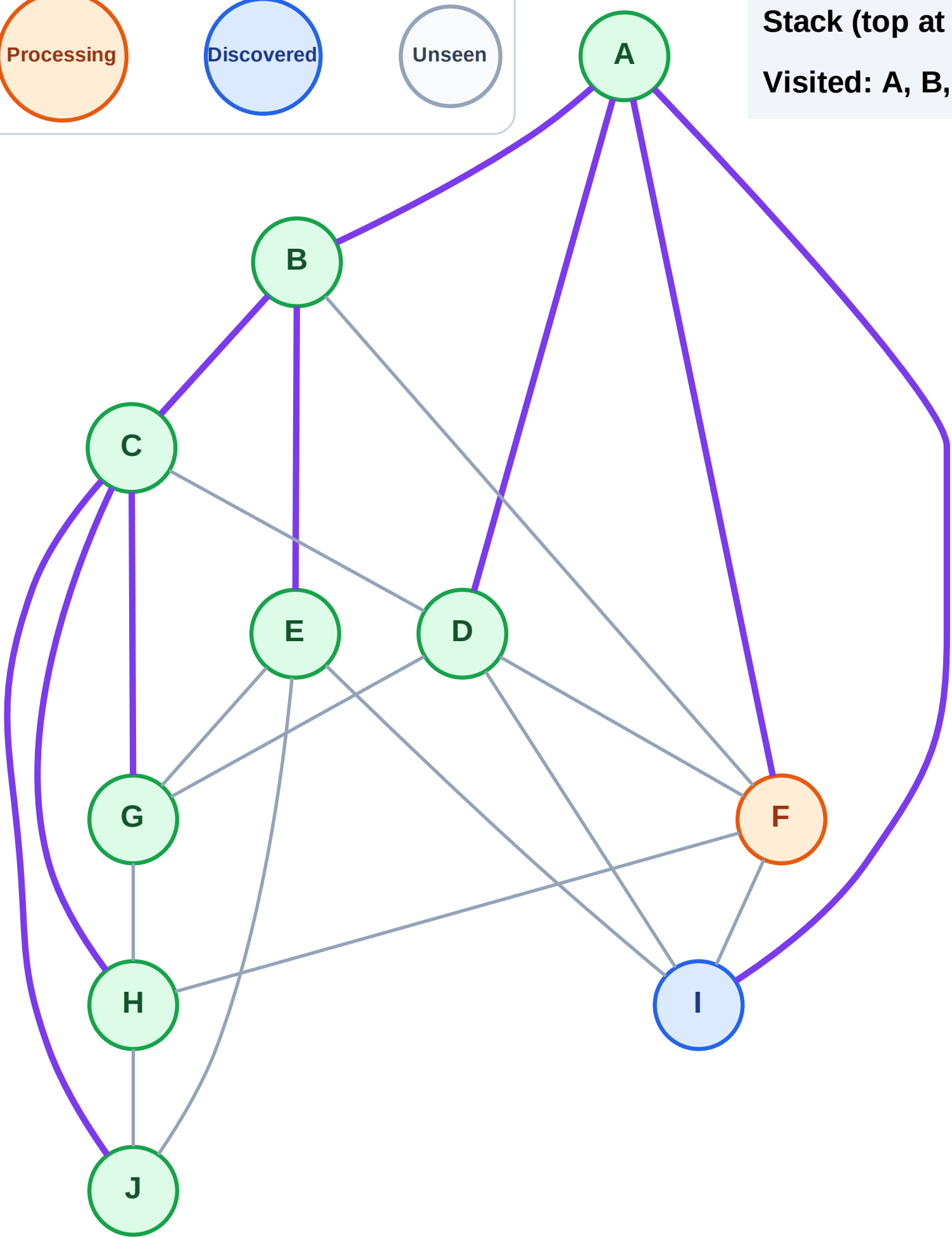
Processing

Discovered

Unseen

Stack (top at right): I

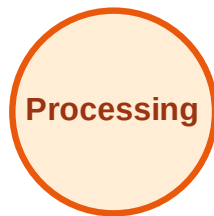
Visited: A, B, C, D, E, G, H, J



DFS Traversal

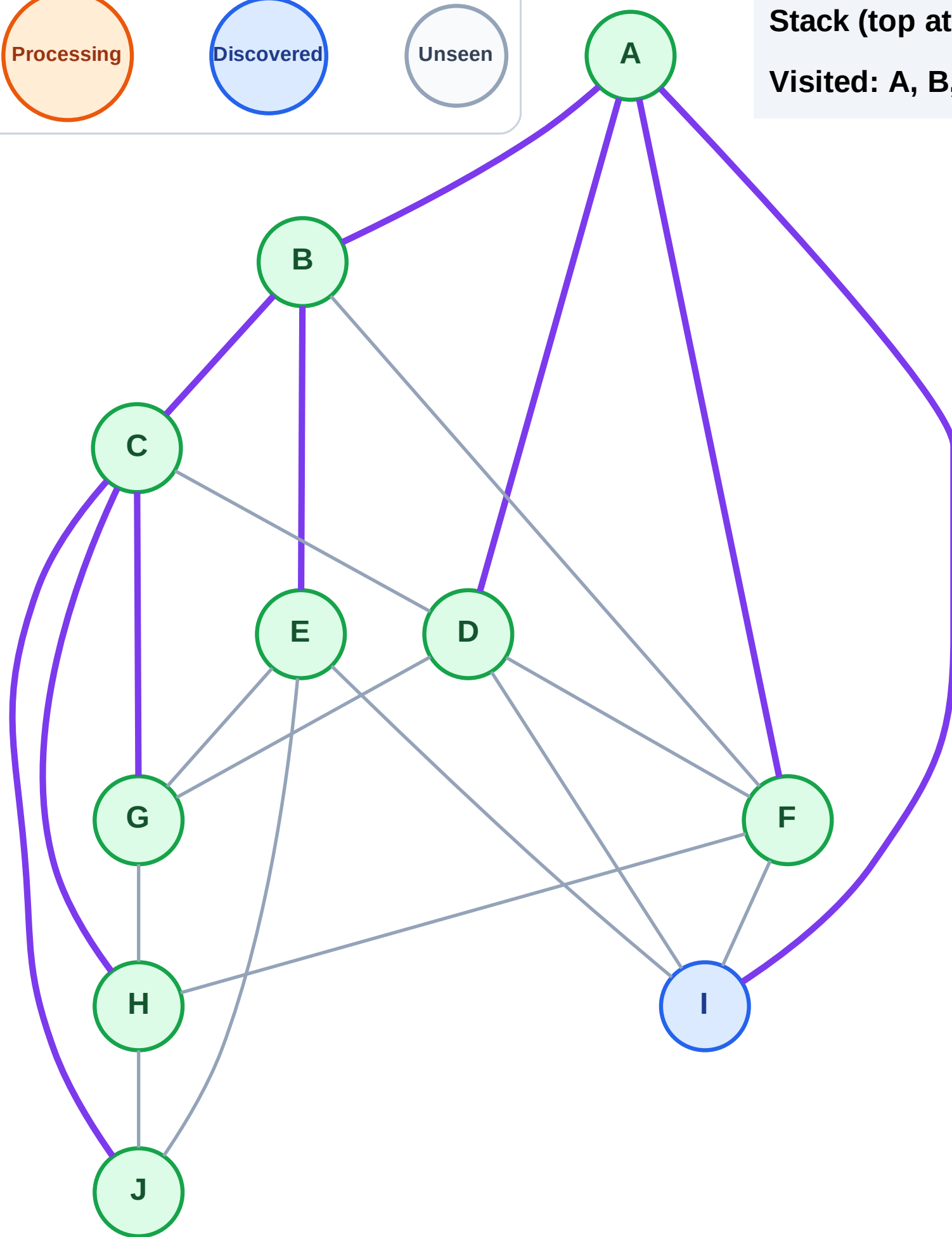
Step 19: No new neighbors from F

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

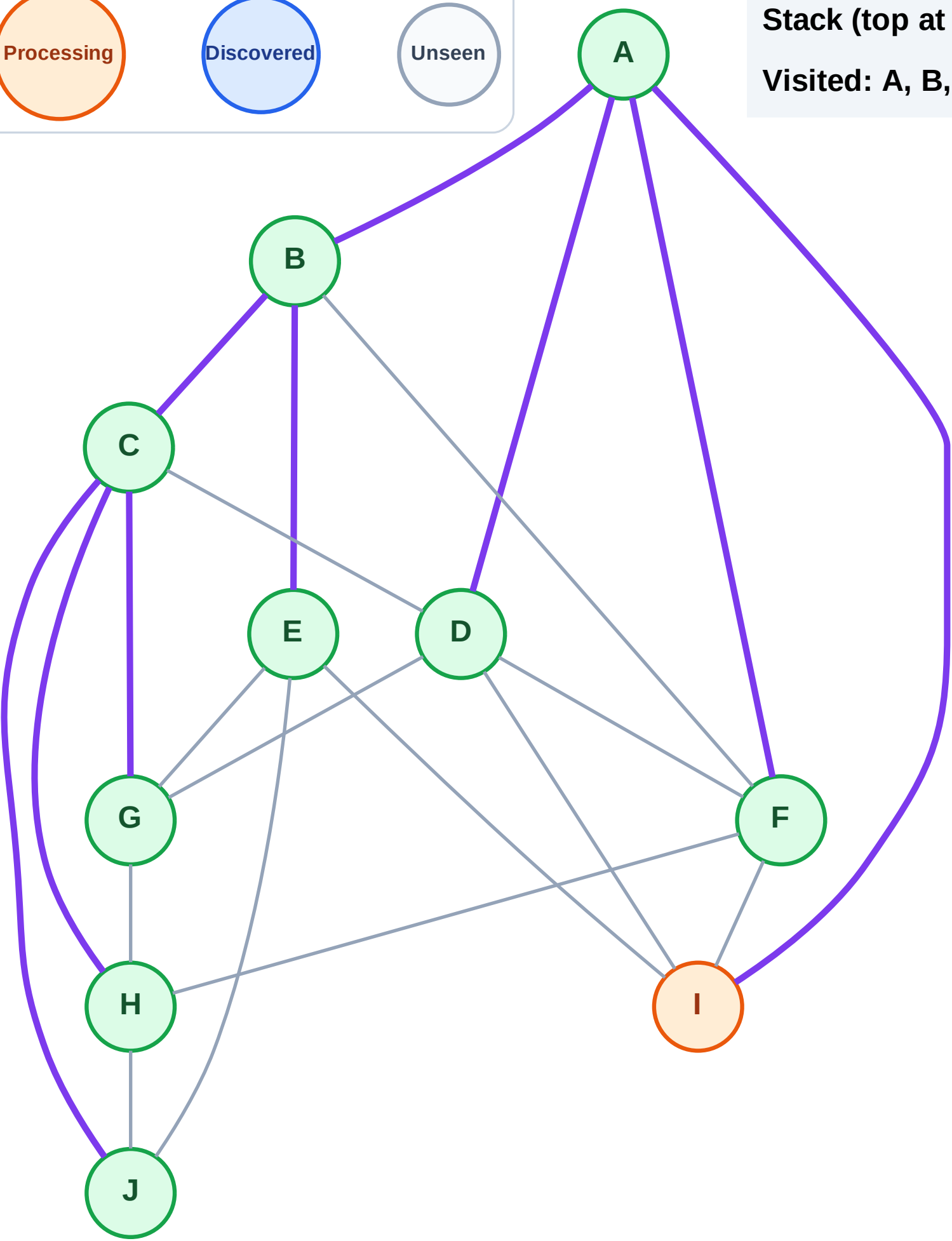
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

